

Chapter 9

Relativistic Mechanics

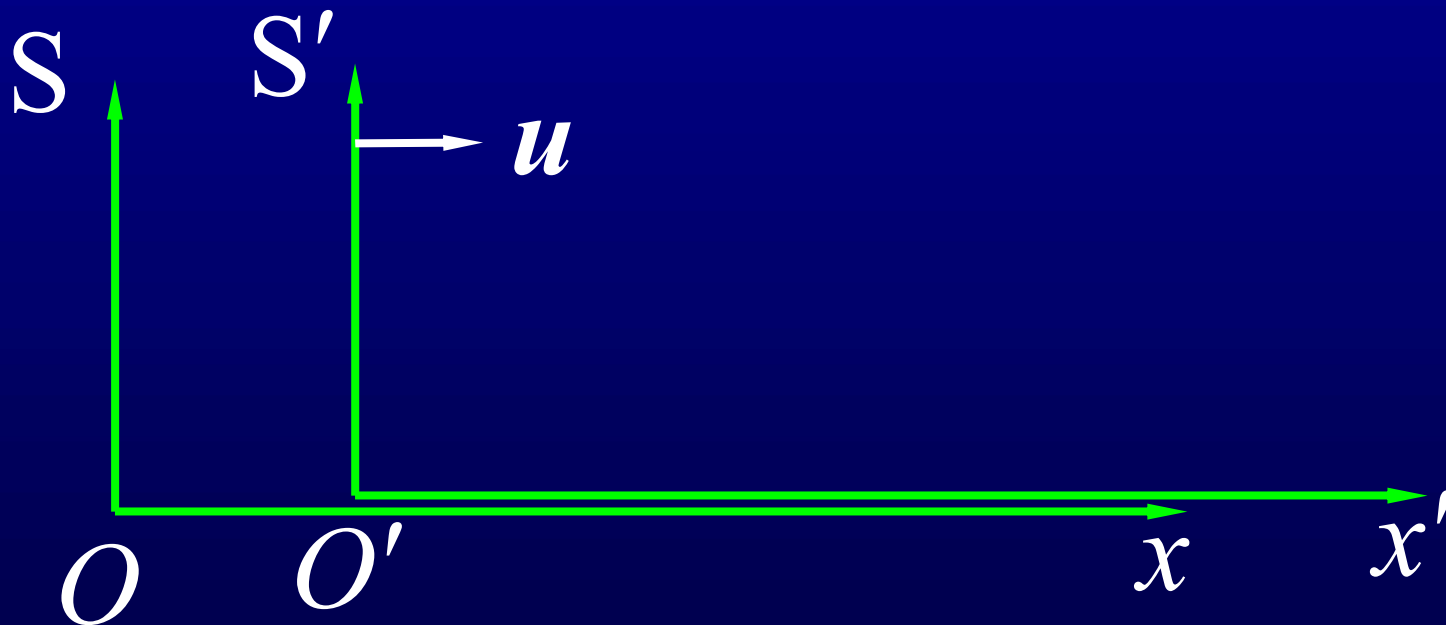
9.1 Galilean transformation

Standard configuration:



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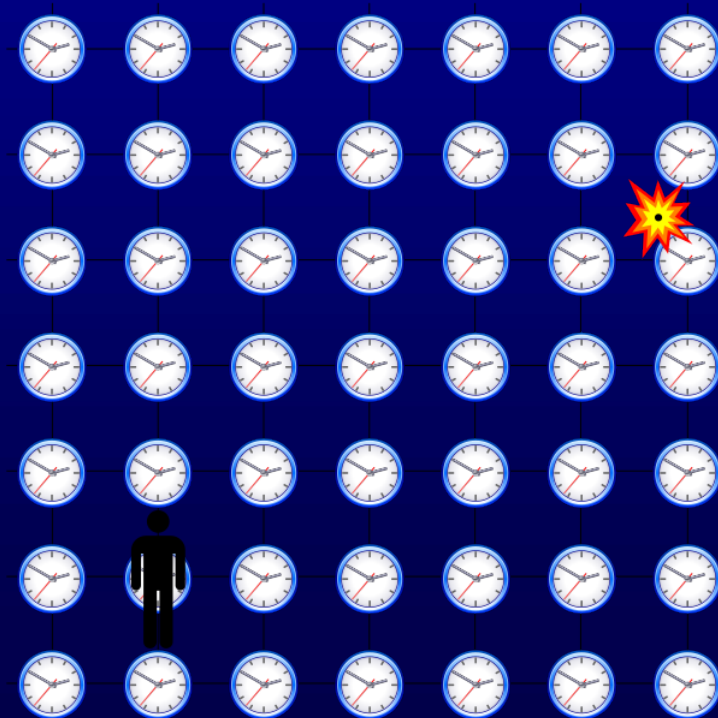


Event: An event is any physical occurrence that happens instantaneously at a single point in spacetime, represented by a set of coordinates (x, y, z, t) .

An event in S (x, y, z, t)

The same event in S' (x', y', z', t')

Observer: The term 'observer' refers to an entire ensemble of synchronized clocks associated with one inertial reference frame.



An event in S

$$(x, y, z, t)$$

The same event in S'

$$(x', y', z', t')$$

Galilean transform:

$$\left\{ \begin{array}{l} x' = x - ut \\ y' = y \\ z' = z \\ t' = t \end{array} \right.$$

Results of Galilean transform

$$t'_A - t'_B = t_A - t_B$$

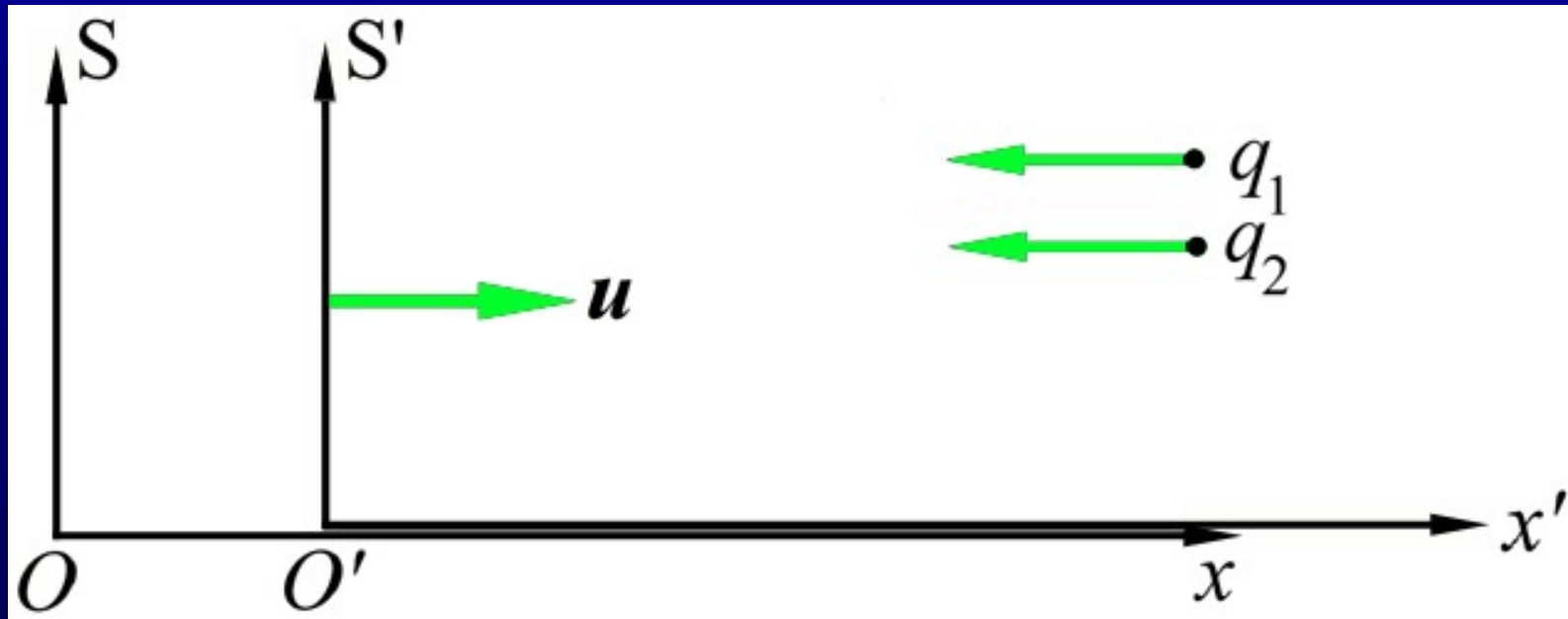
$$\begin{aligned} x'_A - x'_B &= (x_A - ut_A) - (x_B - ut_B) \\ &= x_A - x_B - \underline{u(t_A - t_B)} \end{aligned}$$

$$v'_x = v_x - u$$

$$a'_x = a_x$$

Galileo's relativity: The laws of mechanics are invariant under a Galilean transformation.

q_1, q_2 at rest in S system



In S frame, Coulomb force;

In S', magnetic force comes into play.

Wave equation: $\frac{\partial^2}{\partial x^2} \varphi - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \varphi = 0$

$$x' = x - ut$$

$$\frac{\partial}{\partial x} = \frac{\partial}{\partial x'} \frac{\partial x'}{\partial x};$$

$$\frac{\partial}{\partial t} = \frac{\partial}{\partial x'} \frac{\partial x'}{\partial t} + \frac{\partial}{\partial t'} \frac{\partial t'}{\partial t} = -u \frac{\partial}{\partial x'} + \frac{\partial}{\partial t'}$$

$$\frac{\partial^2}{\partial t^2} = -u \frac{\partial^2}{\partial x'^2} (-u) - u \frac{\partial^2}{\partial x' \partial t'} + \frac{\partial}{\partial t'} \left(-u \frac{\partial}{\partial x'} \right) + \frac{\partial^2}{\partial t'^2}$$

$$\frac{\partial^2}{\partial x'^2} \varphi - \frac{1}{c^2} \frac{\partial^2}{\partial t'^2} \varphi + \frac{2u}{c^2} \frac{\partial^2}{\partial x' \partial t'} \varphi - \frac{u^2}{c^2} \frac{\partial^2}{\partial x'^2} \varphi = 0$$

Ether & Michelson-Morley experiment

Ether is the medium in which light propagates with speed **c** .

If **ether** exists, it should be

- massless (imponderable),
- perfect transparency,
- no resistance to moving objects,
- very stiff (for high speed).

Michelson-Morley interferometer

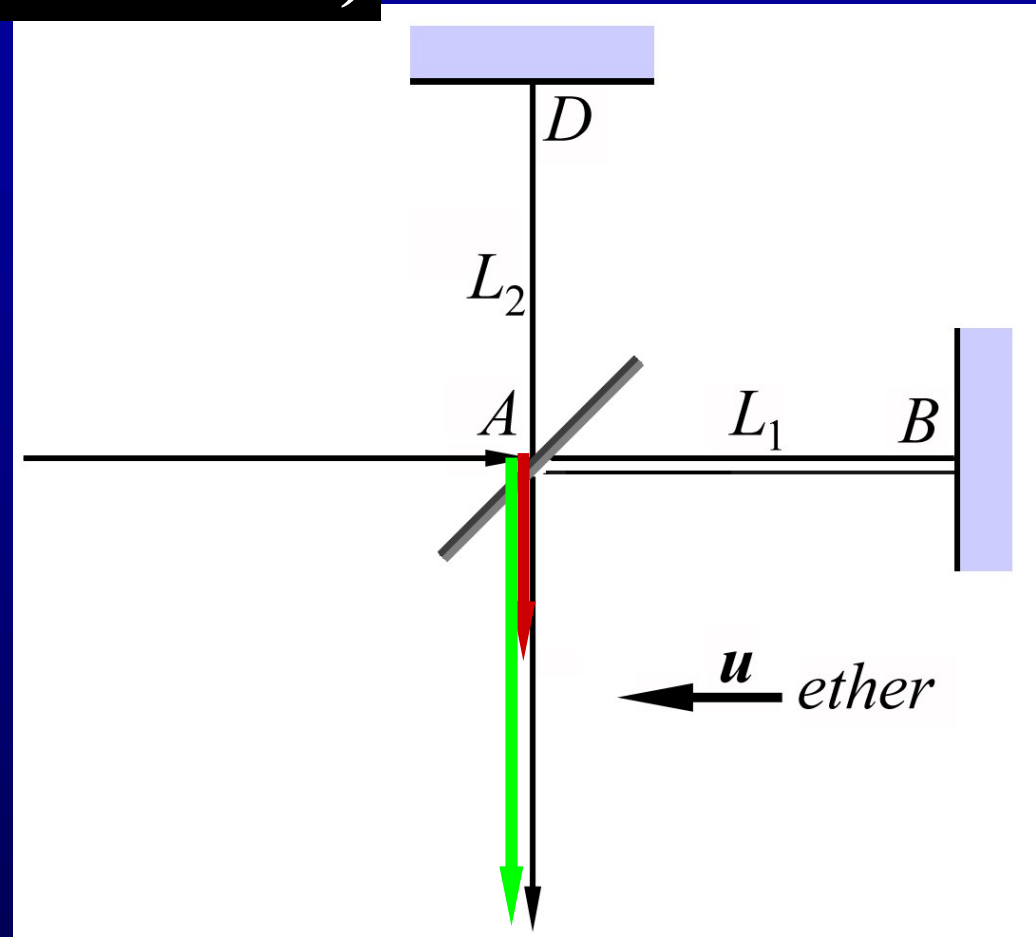


In S' system (interferometer)

$$t_{AB} = \frac{L_1}{c - u}$$

$$t_{BA} = \frac{L_1}{c + u}$$

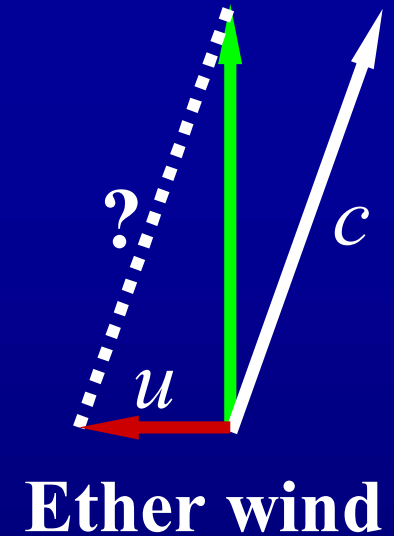
$$t_{ABA} = t_{AB} + t_{BA} = \frac{2L_1}{c} \frac{1}{1 - \beta^2} \left(\beta \equiv \frac{u}{c} \right)$$



In AD direction

$$v_{\perp} = \sqrt{c^2 - u^2}$$

$$t_{ADA} = \frac{2L_2}{\sqrt{c^2 - u^2}} = \frac{2L_2}{c} \frac{1}{\sqrt{1 - \beta^2}}$$



Time difference is

$$\Delta t \equiv t_{ABA} - t_{ADA} = \frac{2L_1}{c} \frac{1}{1 - \beta^2} - \frac{2L_2}{c} \frac{1}{\sqrt{1 - \beta^2}}$$

$$\Delta t \equiv t_{ABA} - t_{ADA} = \frac{2L_1}{c} \frac{1}{1 - \beta^2} - \frac{2L_2}{c} \frac{1}{\sqrt{1 - \beta^2}}$$

Turn the instrument through 90°

$$\Delta t' = t'_{ABA} - t'_{ADA} = \frac{2L_1}{c} \frac{1}{\sqrt{1 - \beta^2}} - \frac{2L_2}{c} \frac{1}{1 - \beta^2}$$

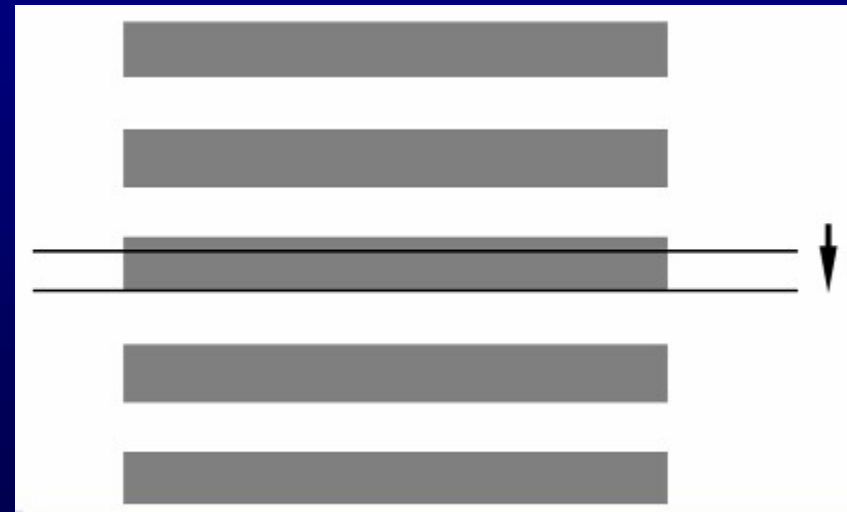
$$\begin{aligned} \Delta t - \Delta t' &= \frac{2(L_1 + L_2)}{c(1 - \beta^2)} - \frac{2(L_1 + L_2)}{c\sqrt{1 - \beta^2}} \\ &\approx \frac{L_1 + L_2}{c} \beta^2 \end{aligned}$$

For $L_1 + L_2 = 22\text{m}$ (8 round trips)

using values of $u = 29.8 \text{ km/s}$ and

**wavelength of yellow (sodium) light $\lambda = 589\text{nm}$
interference fringes shift is**

$$\begin{aligned}\Delta N &= \frac{c(\Delta t - \Delta t')}{\lambda} \\ &= \frac{(L_1 + L_2)\beta^2}{\lambda} \approx 0.40\end{aligned}$$



The result was negative!

- In 1892, G. F. Fitzgerald and H. A. Lorentz independently proposed the concept of *contraction of moving length*.
- Lorentz(1899) proposed that the *time intervals* measured at moving objects would become larger than expected. *Lorentz transformation*
- H. Poincaré(1904) suggested that the *mass* of an object increases as the object's speed increases, he also showed that there is a *maximum speed* which any object can attain—the speed of light.

9.2 Lorentz transformation

Two basic principles of **S**pecial **T**heory of **R**elativity:

(1) *The laws of physics are the same in all inertial systems. No preferred inertial system exists. (the principle of relativity)*

(2) *The speed of light in free space has the same value in all inertial systems. (the principle of the constancy of the speed of light).*

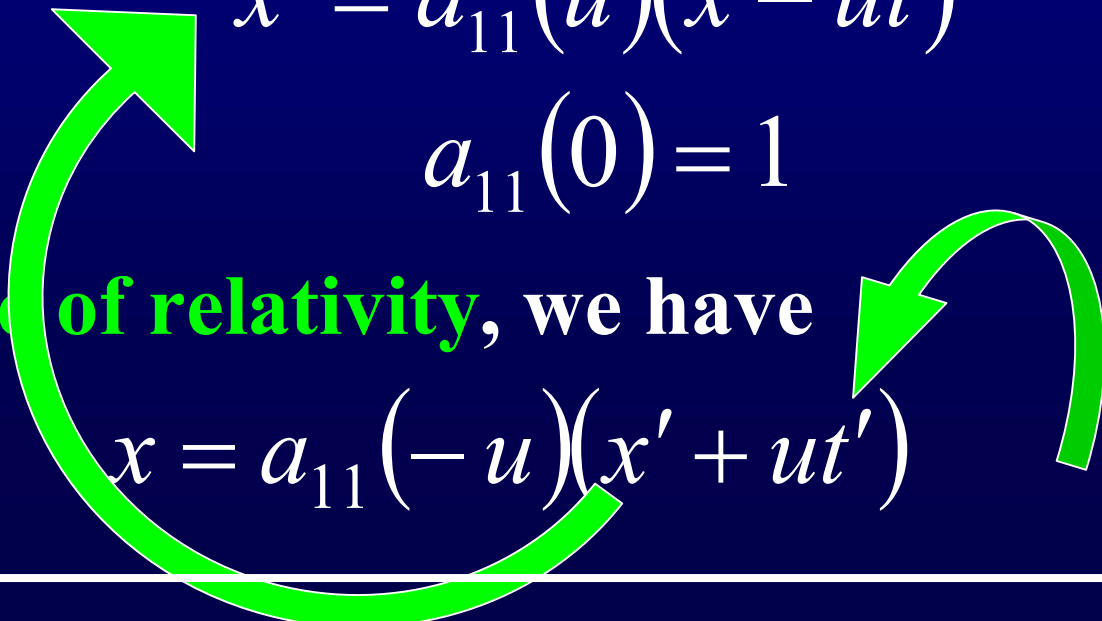
* Derivation of Lorentz transformation

$$\begin{cases} x' = a_{11}x + a_{12}y + a_{13}z + a_{14}t \\ y' = y \\ z' = z \end{cases}$$

For arbitrary y, z , if $x' = 0$ $x = ut$

$$x' = a_{11}(u)(x - ut)$$
$$a_{11}(0) = 1$$

By **the principle of relativity**, we have

$$x = a_{11}(-u)(x' + ut')$$


$$t' = f[x, x'(x, t)] = a_{44}t + a_{41}x$$

$$\begin{cases} x' = a(x - ut) \\ t' = b(t - ex) \end{cases}$$

Solve for x :

$$x = \frac{1}{\Delta} \begin{vmatrix} x' & -au \\ t' & b \end{vmatrix} = \frac{1}{\Delta} (bx' + aut')$$

Compare it with $x = a_{11}(-u)(x' + ut')$

and have $a = b$

the constancy of speed of light

$t = t' = 0$ a spherical light wave is emitted from O, O'

$$x^2 + y^2 + z^2 = c^2 t^2$$

$$x'^2 + y'^2 + z'^2 = c^2 t'^2$$


$$\left\{ \begin{array}{l} x' = a(x - ut) \\ t' = a(t - ex) \end{array} \right.$$

$$\left\{ \begin{array}{l} e = \frac{u}{c^2} \\ a = \pm \frac{1}{\sqrt{1 - \beta^2}} \equiv \pm \gamma \end{array} \right.$$

$$\begin{array}{l} u = 0 \\ x = +x' \end{array}$$

$$x'^2 + y'^2 + z'^2 = c^2 t'^2 \quad \begin{cases} x' = a(x - ut) \\ t' = a(t - ex) \end{cases}$$

$$a^2(x - ut)^2 + y^2 + z^2 = c^2 a^2(t - ex)^2$$

$$a^2(1 - e^2 c^2)x^2 - 2a^2(u - c^2 e)y^2 + z^2 = a^2(c^2 - u^2)t^2$$

$$x^2 + y^2 + z^2 = c^2 t^2$$

$$u - c^2 e = 0 \Rightarrow e = \frac{u}{c^2}.$$

$$a^2(c^2 - u^2) = c^2 \Rightarrow a = \pm \sqrt{\frac{c^2}{c^2 - u^2}}$$

$$= \pm \frac{1}{\sqrt{1 - \beta^2}} \equiv \pm \gamma, \quad \beta \equiv \frac{u}{c}.$$

$$a^2(1 - e^2 c^2) = \frac{1}{1 - \beta^2} \left(1 - \frac{u^2}{c^4} c^2 \right) = 1.$$

Lorentz Transformation

$$\begin{cases} x' = \gamma (x - ut) \\ y' = y \\ z' = z \\ t' = \gamma \left(t - \frac{u}{c^2} x \right) \end{cases}$$

$$\gamma = \frac{1}{\sqrt{1 - \beta^2}} \quad \beta = \frac{u}{c}$$

Lorentz transformation
for event **pair** (x'_1, t'_1) (x'_2, t'_2)

$$\left\{ \begin{array}{l} \Delta x' = \gamma (\Delta x - u \Delta t) \\ \Delta y' = \Delta y \\ \Delta z' = \Delta z \\ \Delta t' = \gamma \left(\Delta t - \frac{u}{c^2} \Delta x \right) \end{array} \right.$$

- $u \leq c, \quad \gamma \equiv \frac{1}{\sqrt{1 - \beta^2}}; \quad \beta \equiv \frac{u}{c}$

- **Inverse transformation**

- **When $u \ll c$ it is equivalent to Galilean transformation**

- **invariance**

$$\begin{aligned} (\Delta s)^2 &= (c\Delta t)^2 - (\Delta x)^2 - (\Delta y)^2 - (\Delta z)^2 \\ &= (c\Delta t')^2 - (\Delta x')^2 - (\Delta y')^2 - (\Delta z')^2 = (\Delta s')^2 \end{aligned}$$

Δs is **spacetime interval** of the event pair.

When $(c\Delta t)^2 > (\Delta x)^2$

$$\Delta\tau \equiv \sqrt{(\Delta t)^2 - (\Delta x/c)^2}$$

**proper time interval
of the event pair.**

When $(c\Delta t)^2 < (\Delta x)^2$

$$\Delta\sigma = \sqrt{(\Delta x)^2 - (c\Delta t)^2}$$

**proper space
interval.**

When $(c\Delta t)^2 = (\Delta x)^2$

$$\Delta s = 0$$

**can be connected
by a light signal**

**Simultaneous measurement $\Delta x' = \ell_0$ in S'
($\Delta t' = 0$) constructs a pair of events.**

The proper distance is $\Delta\sigma = \sqrt{(\Delta x')^2 - (c\underline{\Delta t'})^2} = \ell_0$

For **same pair of events in S system**

$$\Delta x = \gamma(\Delta x' + u\Delta t') = \gamma\Delta x' = \gamma\ell_0 \leftarrow$$

$$\Delta t = \gamma\left(\Delta t' + \frac{u}{c^2}\Delta x'\right) = \gamma\frac{u}{c^2}\Delta x' \neq 0$$

But

$$\Delta\sigma = \sqrt{(\gamma\Delta x')^2 - \left(c\gamma\frac{u}{c^2}\Delta x'\right)^2} = \gamma\ell_0\sqrt{1 - \beta^2} = \ell_0$$

Time dilation (moving clocks run slow)

In S' , consider a clock at $x' = 0$.

Two readings form a pair events of $\Delta x' = 0$, $\Delta t'$

The time interval of same pair in S is

$$\Delta t = \gamma \left(\Delta t' + \frac{u}{c^2} \cancel{\Delta x'} \right) > \Delta t'$$

There are direct evidences

Lorentz contraction (moving rods contract)

$$x'_2 - x'_1 = \Delta x' = \ell_0 \quad \text{in } S'$$

In S, **simultaneous measurement** of this length yields

$$\Delta t = 0, \Delta x = \Delta x' / \gamma \quad \boxed{\Delta x' = \gamma(\Delta x - u\Delta t)}$$

$$\ell = \ell_0 \sqrt{1 - \frac{v^2}{c^2}}$$

***Can a rod of 20m be **in** a tunnel of 10m, especially viewing from rod-frame?**

Example Mouns are produced with very high speeds in the upper atmosphere when cosmic rays collide with air molecules. The mean (proper) lifetime of a muon is $2.2\mu s$. Take the height of the atmosphere to be 100km. Find the minimum speed of the mouns that will enable them to survive the journey to the surface of the earth.

Solution: In the earth's frame of reference. The time necessary for the muon to travel from the top of the atmosphere to the surface of the earth is

$$\Delta t = \frac{L_0}{c} = \frac{100km}{3.0 \times 10^8 m/s} = 333 \mu s.$$

Because of the time dilation, we have

$$\Delta t = 333 \mu m = \frac{\Delta t_0}{\sqrt{1 - \frac{u^2}{c^2}}} = \frac{2.2 \mu m}{\sqrt{1 - \frac{u^2}{c^2}}}.$$

Solving the above equation, we find

$$u = 0.999978c.$$

In the muon's frame of reference, the distance the muon can travel is no greater than

$$L = c\Delta t_0 = 660m.$$

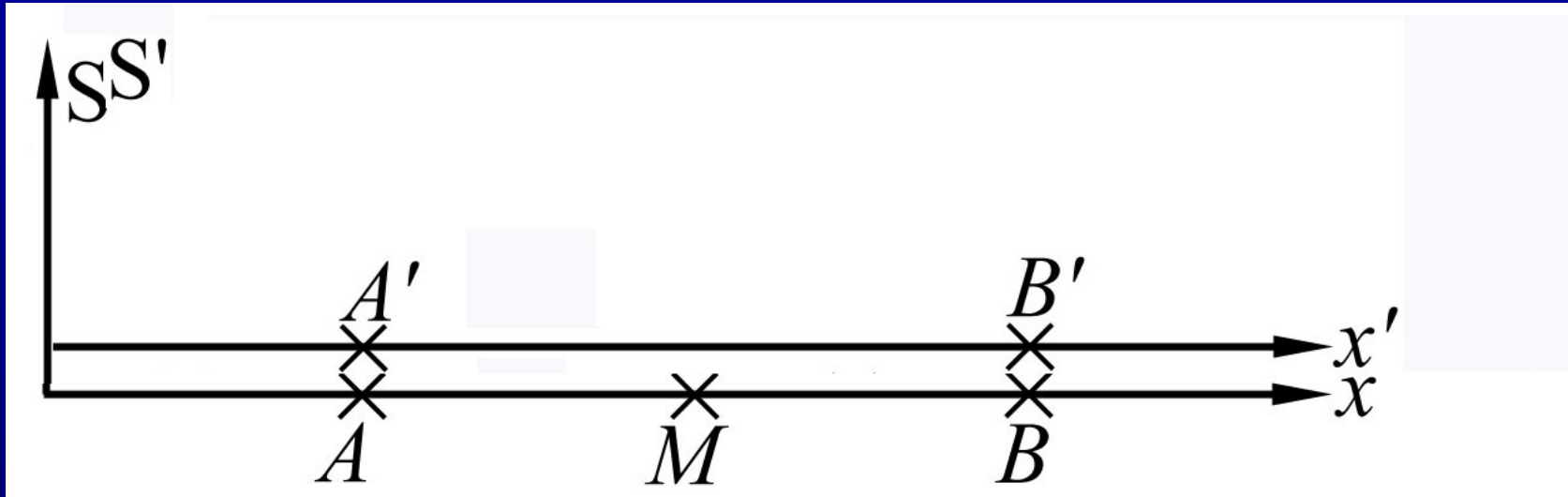
Because of the length contraction, we have

$$L = 660m = L_0 \sqrt{1 - \frac{u^2}{c^2}} = (100km) \sqrt{1 - \frac{u^2}{c^2}}.$$

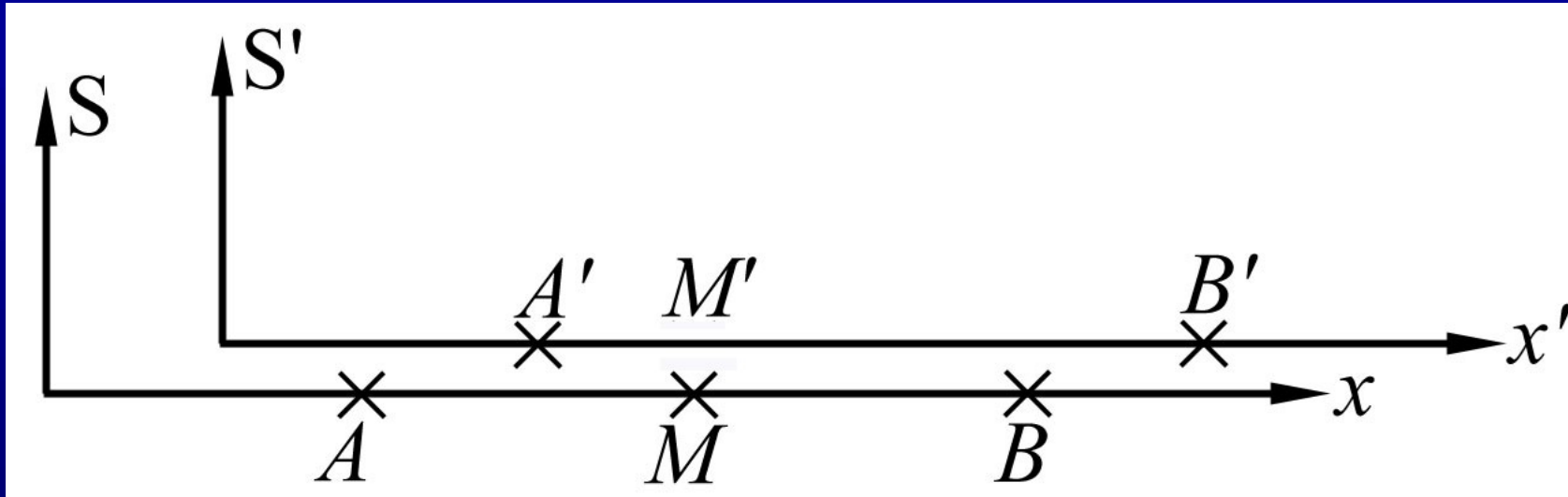
Solving the above equation, we find

$$u = 0.999978c.$$

Relativity of simultaneity



Relativity of simultaneity



$$t'_B - t'_A = -\gamma \frac{u}{c^2} (x_B - x_A) \quad \boxed{\Delta t = 0 \text{ implied}}$$

$$\Delta x = 0 \longrightarrow \Delta t' = 0$$

* Causality and signal speed

An event $P(x_P, t_P)$ causes an event $Q(x_Q, t_Q)$

Signal (information or interaction or ...) is sent at speed:

$$v_S = \frac{x_Q - x_P}{t_Q - t_P} = \frac{\Delta x}{\Delta t}$$

If P and Q are uncorrelated, v_S is only a quantity of dimension LT^{-1} , any value being allowed.

An observer in $S'(u)$ ($u < c$)

$$\begin{aligned}\Delta t' &= \gamma \left(\Delta t - \frac{u}{c^2} \Delta x \right) = \gamma \Delta t \left(1 - \frac{u}{c^2} \frac{\Delta x}{\Delta t} \right) \\ &= \gamma \Delta t \left(1 - \frac{uv_S}{c^2} \right)\end{aligned}$$

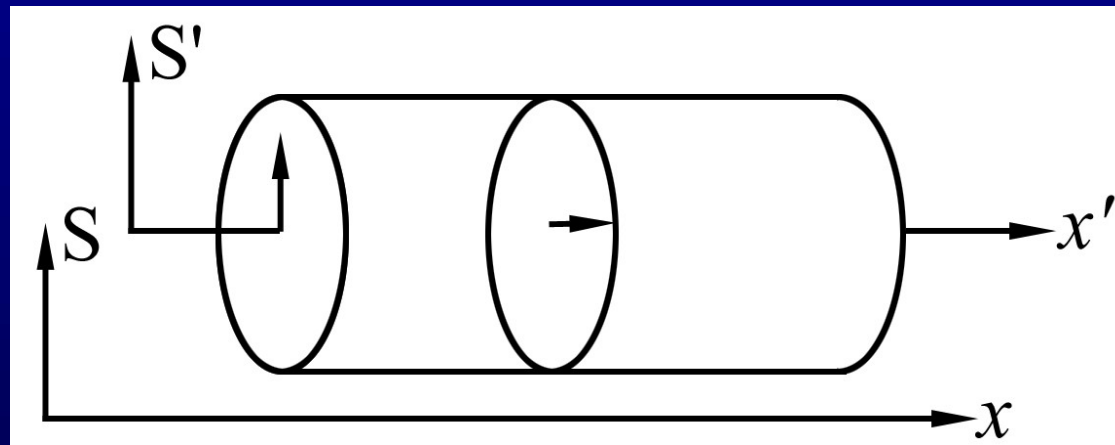
For causal events, time sequence can not change, $v_S \leq c$.

For uncorrelated events, v_S may be greater than c . Time sequence may change if

$$c > u > (c/v_S)c$$

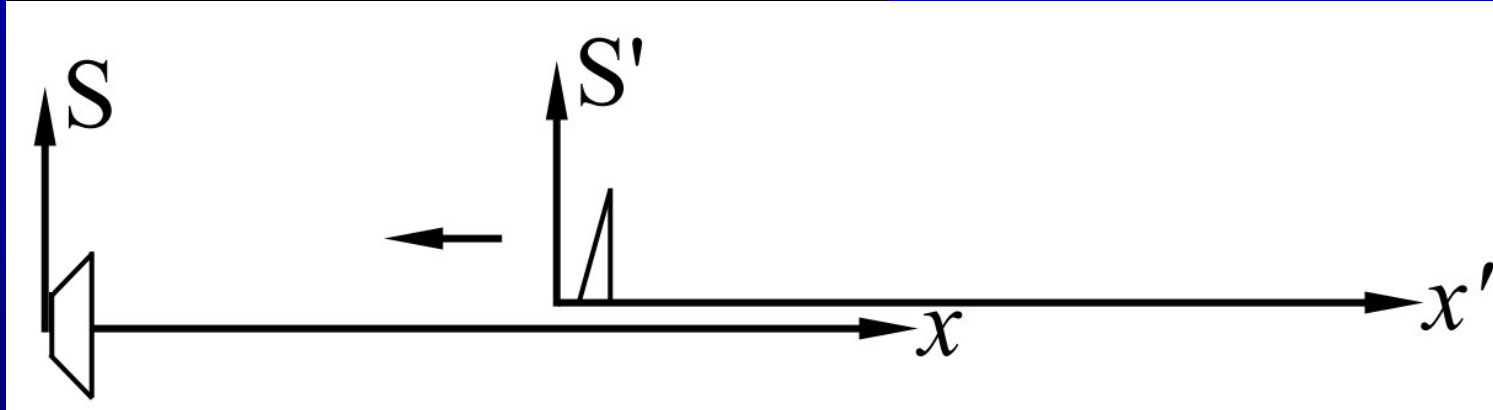
* Laue's cylinder

Relativistically, “rigid bodies” and “incompressible fluids” are impossible, for a rigid body can transmit signal **instantaneously**.



$$\Delta t' = 0 \quad \Delta x' \neq 0 \quad \Delta t = \gamma \frac{u}{c^2} \Delta x'$$

Optical Doppler effect



In S' two signals are emitted at $(0, 0)$, and $(0, t')$.

In S , they are events $(0,0)$ and (x, t) .

The second signal is issued at (according to observer in S)

$$t = \gamma \left(t' + \frac{u}{c^2} x' \right) = \gamma t'$$

Since then it **propagates** and is received at

$$\begin{aligned} t_1 &= t + x/c = t + ut/c = (1 + \beta)t \\ &= (1 + \beta)\gamma t' \end{aligned}$$

$$\Delta t = (1 + \beta)\gamma\Delta t' = \sqrt{\frac{1 + \beta}{1 - \beta}}\Delta t'$$

If $\Delta t'$, Δt are period, we have

$$\nu_o = \sqrt{\frac{1 - \beta}{1 + \beta}}\nu_e \quad (\text{redshift})$$

For approaching, $\beta \rightarrow -\beta$

$$\nu_o = \sqrt{\frac{1 + \beta}{1 - \beta}}\nu_e \quad (\text{blueshift})$$

		$\lambda_0(\text{nm})$	$\nu(\text{THz})$
Microwaves		$>1\text{cm} - 30\text{m}$	$10^9\text{Hz} - 3 \times 10^{11}\text{Hz}$
Infrared	near	780 – 3000	$3 \times 10^{-1} - 4 \times 10^2$
	intermediate	3000 – 6000	
	far	6000 – 15000	
	extreme	15000nm – 1.0mm	
Red		780– 622	384–482
Orange		622–597	482–503
Yellow		597–577	503–520
Green		577 –492	520–610
Blue		492–455	610–659
Violet		455–390	659–769
Ultraviolet		$3.2 - 1.2 \times 10^3 \text{eV}$	$\sim 8 \times 10^2 - 3 \times 10^5$
X rays		$1.2 \times 10^3 - 2.1 \times 10^5 \text{eV}$	$\sim 3 \times 10^5 - 5 \times 10^7$
Gamma rays		$10^4 - 10^{19} \text{eV}$	

speeding 622nm-577nm

$$\lambda_o = \sqrt{\frac{1-\beta}{1+\beta}} \lambda_e \quad (1-\beta) \left(\frac{\lambda_e}{\lambda_o} \right)^2 = 1 + \beta$$

$$\left(\frac{\lambda_e}{\lambda_o} \right)^2 - 1 = \beta \left[1 + \left(\frac{\lambda_e}{\lambda_o} \right)^2 \right]$$

$$\beta = \frac{\lambda_e^2 - \lambda_o^2}{\lambda_e^2 + \lambda_o^2} = \frac{622^2 - 577^2}{622^2 + 577^2} = 0.0750$$

$$v = 2.25 \times 10^4 \text{ km/s} = 8.10 \times 10^7 \text{ km/h}$$

Lorentz invariance of the wave equation

The wave equation

$$\frac{\partial^2}{\partial x^2} \varphi - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \varphi = 0$$

$$\frac{\partial}{\partial x} = \frac{\partial x'}{\partial x} \frac{\partial}{\partial x'} + \frac{\partial t'}{\partial x} \frac{\partial}{\partial t'}$$

$$\frac{\partial}{\partial t} = \frac{\partial x'}{\partial t} \frac{\partial}{\partial x'} + \frac{\partial t'}{\partial t} \frac{\partial}{\partial t'}$$

$$x' = \gamma (x - ut)$$

$$y' = y$$

$$z' = z$$

$$t' = \gamma \left(t - \frac{u}{c^2} x \right)$$



$$\frac{\partial}{\partial x} = \gamma \frac{\partial}{\partial x'} - \gamma \frac{u}{c^2} \frac{\partial}{\partial t'}$$

$$\frac{\partial}{\partial t} = \gamma \frac{\partial}{\partial t'} - \gamma u \frac{\partial}{\partial x'}$$

$$\frac{\partial^2}{\partial x^2} = \gamma^2 \frac{\partial^2}{\partial x'^2} - \frac{2u}{c^2} \gamma^2 \frac{\partial^2}{\partial x' \partial t'} + \frac{u^2}{c^4} \gamma^2 \frac{\partial^2}{\partial t'^2}$$

$$\frac{\partial^2}{\partial t^2} = \gamma^2 \frac{\partial^2}{\partial t'^2} - 2u \gamma^2 \frac{\partial^2}{\partial x' \partial t'} + u^2 \gamma^2 \frac{\partial^2}{\partial x'^2}$$

$$\begin{aligned}
& \frac{\partial^2}{\partial x'^2} \varphi - \frac{1}{c^2} \frac{\partial^2}{\partial t'^2} \varphi \\
&= \gamma^2 \left(\frac{\partial^2}{\partial x'^2} - \frac{2u}{c^2} \frac{\partial^2}{\partial x' \partial t'} + \frac{u^2}{c^4} \frac{\partial^2}{\partial t'^2} \right) \varphi \\
&\quad - \frac{\gamma^2}{c^2} \left(\frac{\partial^2}{\partial t'^2} - 2u \frac{\partial^2}{\partial x' \partial t'} + u^2 \frac{\partial^2}{\partial x'^2} \right) \varphi \\
&= \gamma^2 \left(1 - \frac{u^2}{c^2} \right) \frac{\partial^2 \varphi}{\partial x'^2} - \frac{1}{c^2} \gamma^2 \left(1 - \frac{u^2}{c^2} \right) \frac{\partial^2 \varphi}{\partial t'^2} \\
&= \frac{\partial^2 \varphi}{\partial x'^2} - \frac{1}{c^2} \frac{\partial^2 \varphi}{\partial t'^2} = 0
\end{aligned}$$

Assignment: 9.2, 9.3, 9.6

When Einstein was 16, in 1895, he asked himself an interesting question:

"If I pursue a beam of light with the velocity c I should observe such a beam of light as a spatially oscillatory electromagnetic field at rest. However, there seems to be no such thing, whether on the basis of experience or according to [the theory of electricity and magnetism]. From the very beginning it appeared to me intuitively clear that, judged from the standpoint of such an observer, everything would have to happen according to the same laws as for an observer who, relative to the earth, was at rest. For how, otherwise, should the first observer know, i.e., be able to determine, that he is in a state of uniform motion?" -- As later written by Einstein in "Autobiographical Notes", in Schilpp, ed., **Albert Einstein: Philosopher-Scientist**.

He continued to work on this question for 10 years with the mixture of concentration and determination that characterised much of his work. He published his answer in 1905:

"... light is always propagated in empty space with a definite velocity c which is independent of the state of [relative] motion of the emitting body The introduction of a 'luminiferous ether' will be superfluous inasmuch as the view here to be developed will not require an 'absolutely stationary space' provided with special properties." -- Annalen Physik 17 (1905).

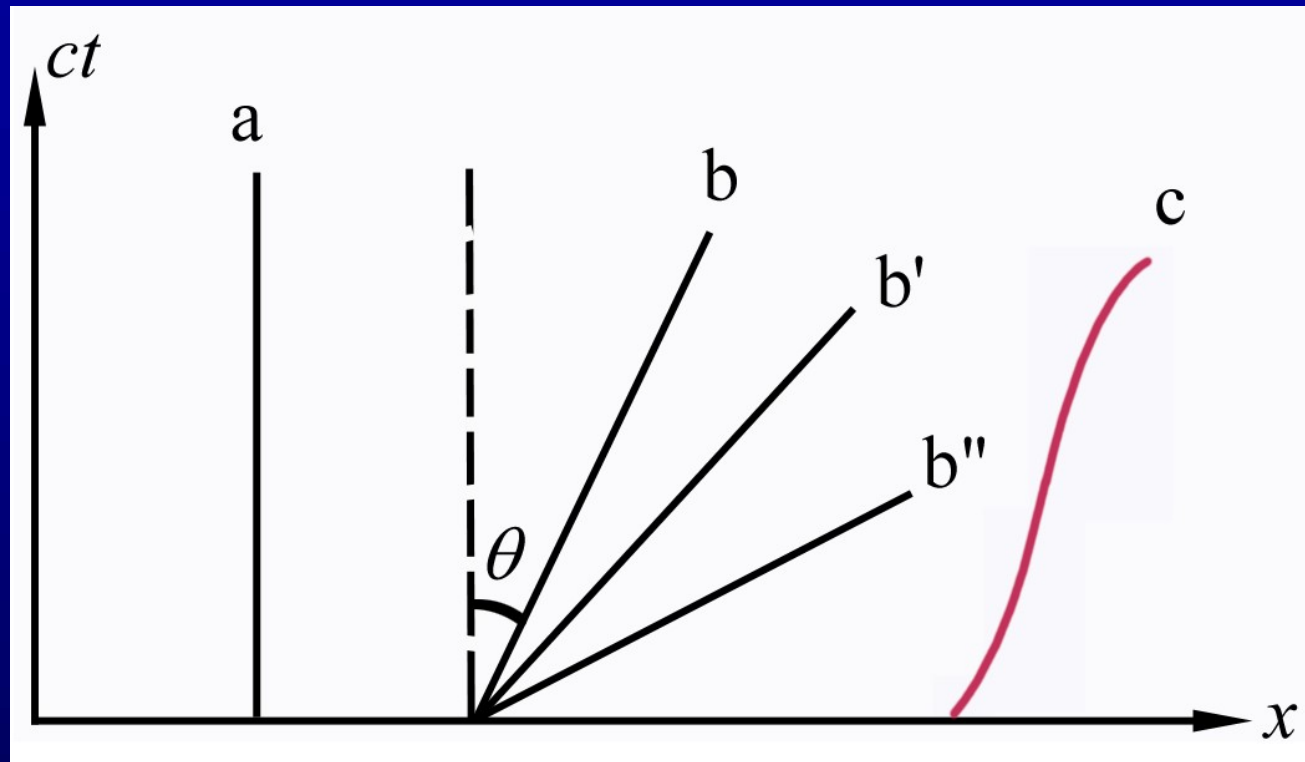
“The special theory of relativity owes its origins to Maxwell’s equations of the electromagnetic field.”

Albert Einstein, *Autobiographical Notes*, p. 59

Walter Isaacson Einstein: His Life and Universe

Chapter One The Light-beam Rider

* 9.3 Spacetime diagram and twin paradox



For b $ct = \cot \theta \ x$

$$\tan \theta = \frac{1}{c} \frac{dx}{dt} = \frac{u}{c} = \beta < 1$$

tachyon(快子)

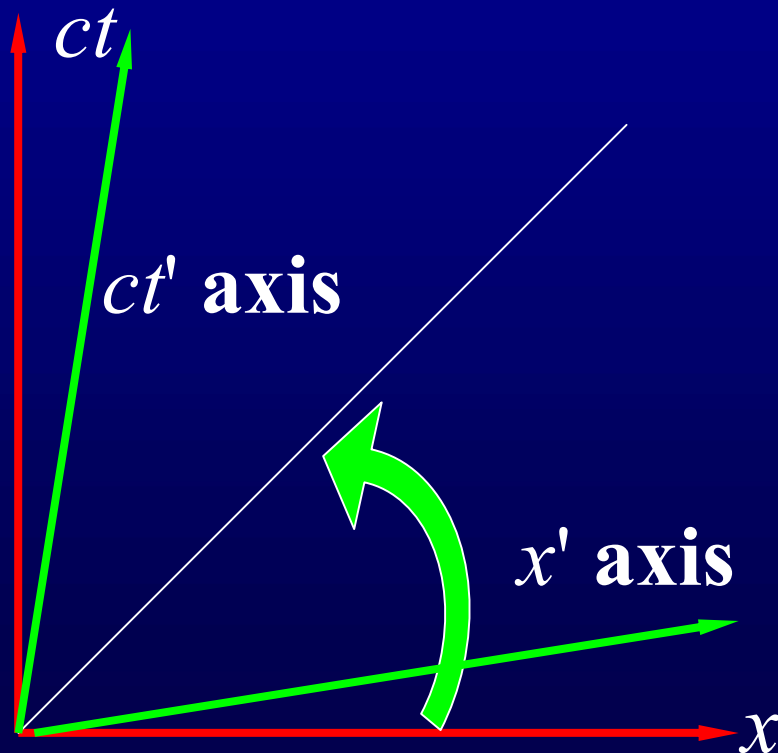
bradyon

To describe events in S'

The axes ct' and x' are lines $x' = 0$ and $ct' = 0$.

$$x = \beta ct$$

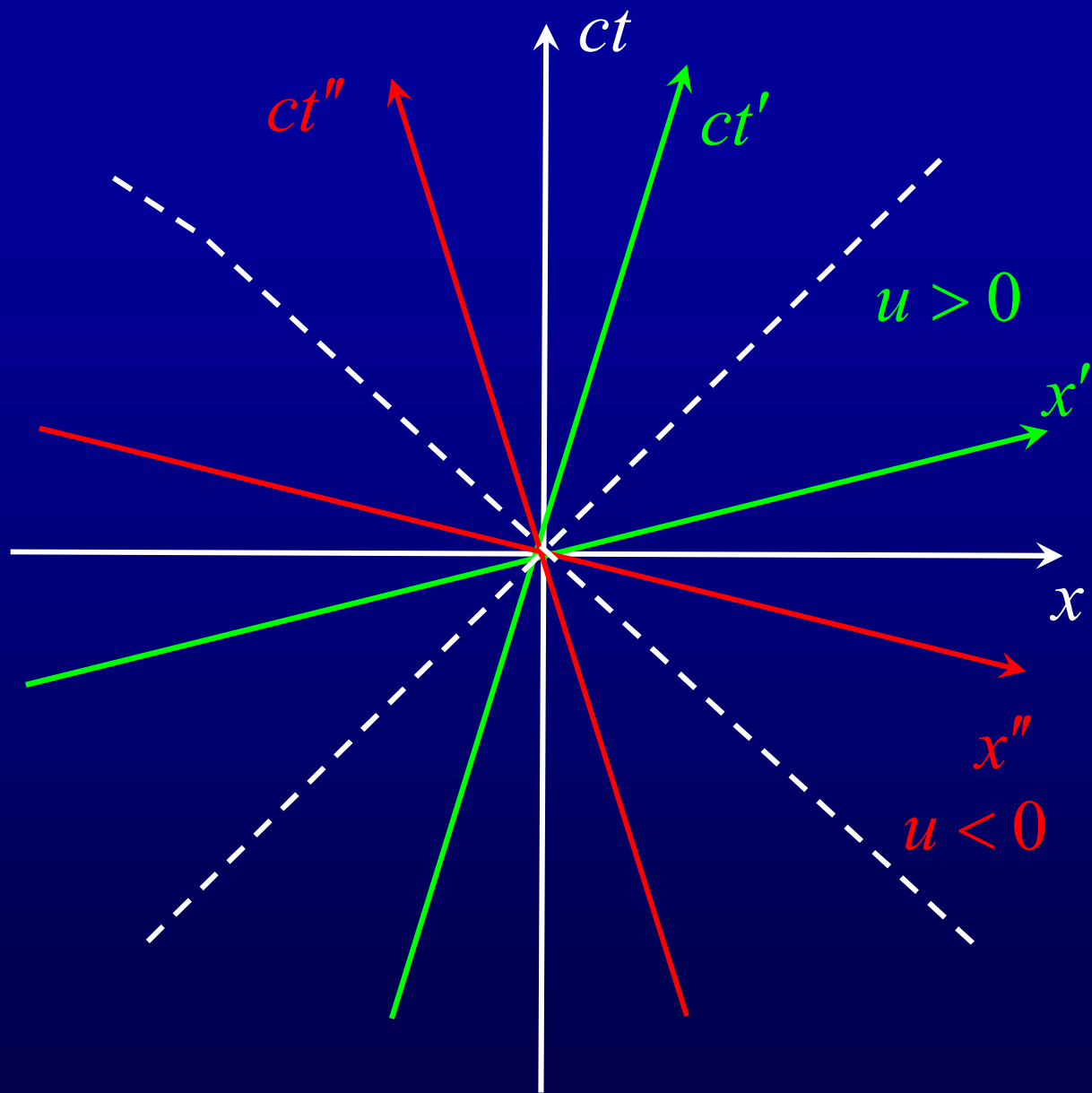
$$ct = \beta x$$



from

$$x' = \gamma (x - ut)$$

$$t' = \gamma \left(t - \frac{u}{c^2} x \right)$$



Unit in S'

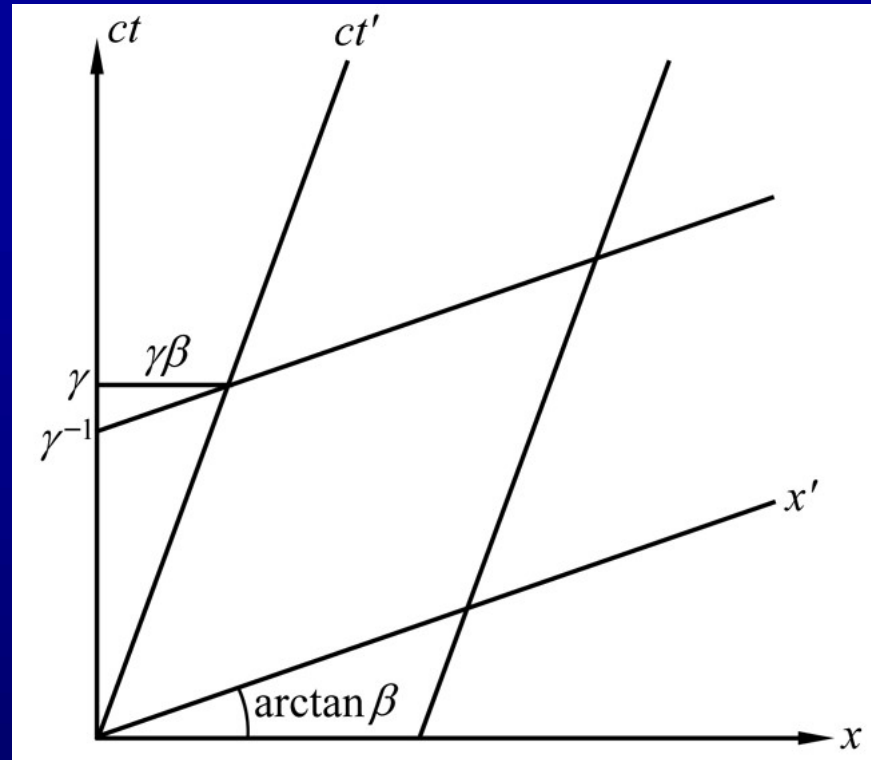
$$(x' = 0, ct' = 1)$$

$$(x' = 1, ct' = 0)$$

i.e.

$$\begin{cases} 0 = x - \beta ct \\ 1 = \gamma (ct - \beta x) \end{cases}$$

$$\begin{cases} 1 = \gamma (x - \beta ct) \\ 0 = ct - \beta x \end{cases}$$



$$1 = \gamma (ct - \beta^2 ct)$$

$$ct = \frac{1}{\gamma(1 - \beta^2)} = \gamma$$

$$1 = \gamma \left(\frac{x}{\beta} - \beta x \right)$$

$$x = \frac{1}{\gamma \left(\frac{1}{\beta} - \beta \right)} = \frac{\beta}{\sqrt{1 - \beta^2}} = \gamma \beta$$

$$\begin{cases} 0 = x - \beta ct \\ 1 = \gamma (ct - \beta x) \end{cases}$$

$$l' \times \cos \theta = \gamma$$

$$l' = \frac{\gamma}{\cos \theta} = \frac{\gamma}{\cos(\arctan \beta)}$$

$$= \gamma \sqrt{1 + \tan^2(\arctan \beta)}$$

$$= \frac{\sqrt{1 + \beta^2}}{\sqrt{1 - \beta^2}}$$

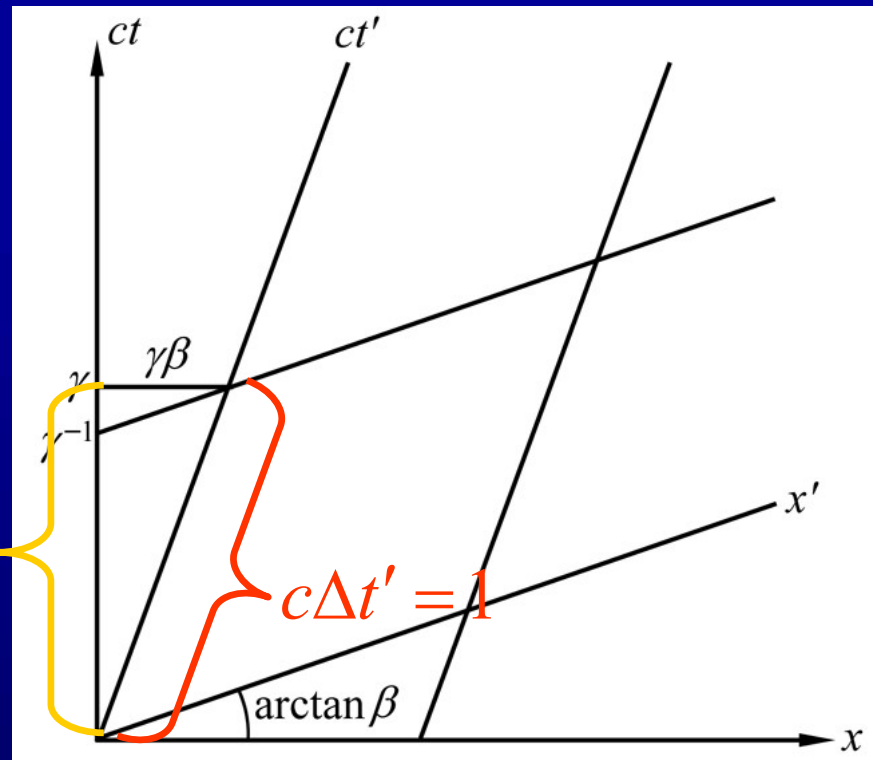
$$\frac{1}{\cos^2 \theta} = \frac{\cos^2 \theta + \sin^2 \theta}{\cos^2 \theta} = 1 + \tan^2 \theta$$

Time Dilation

$$\Delta t' = \frac{1}{c}$$

$$\Delta t = \frac{\gamma}{c} = \gamma \Delta t'$$

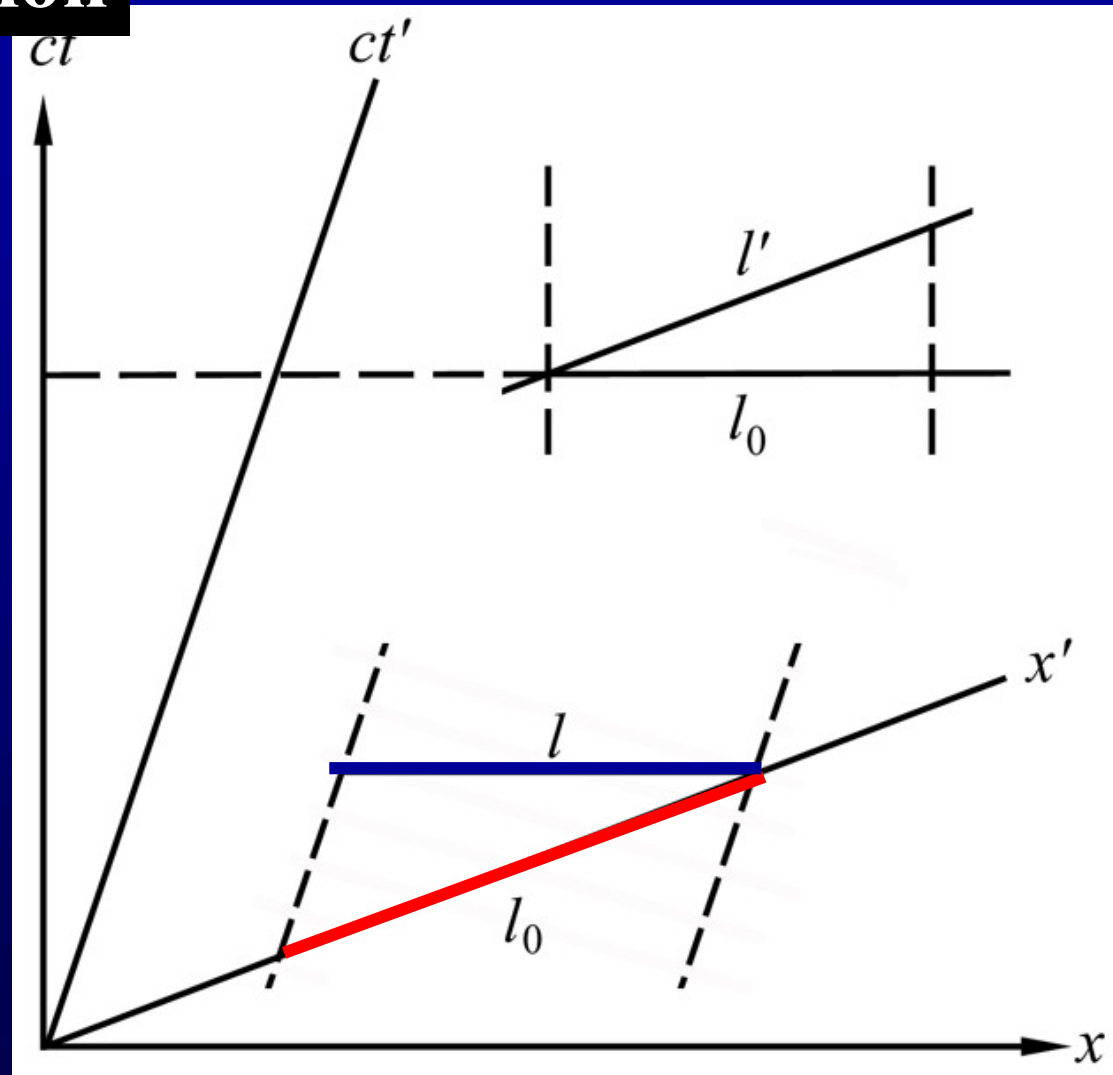
$$c\Delta t = \gamma$$



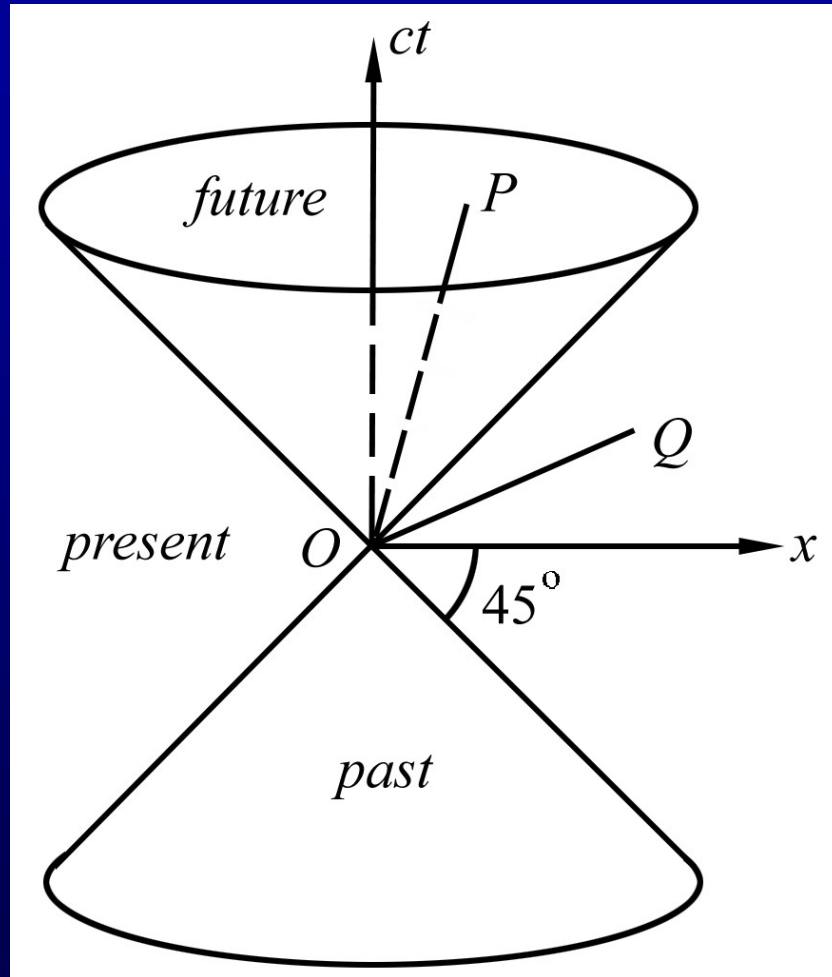
$$\left(\gamma = \frac{1}{\sqrt{1 - \beta^2}} \right)$$

Lorentz contraction

$$\begin{aligned}
 l' &= \frac{l_0}{\gamma} \\
 &= \frac{l_0 \cos \theta}{\gamma \cos \theta} = l_0
 \end{aligned}$$



Light cone



Events pair OP
may be **on some t' axis**
— same x' , at different
time sequence

called timelike pair

All event pairs inside
light cone are timelike.

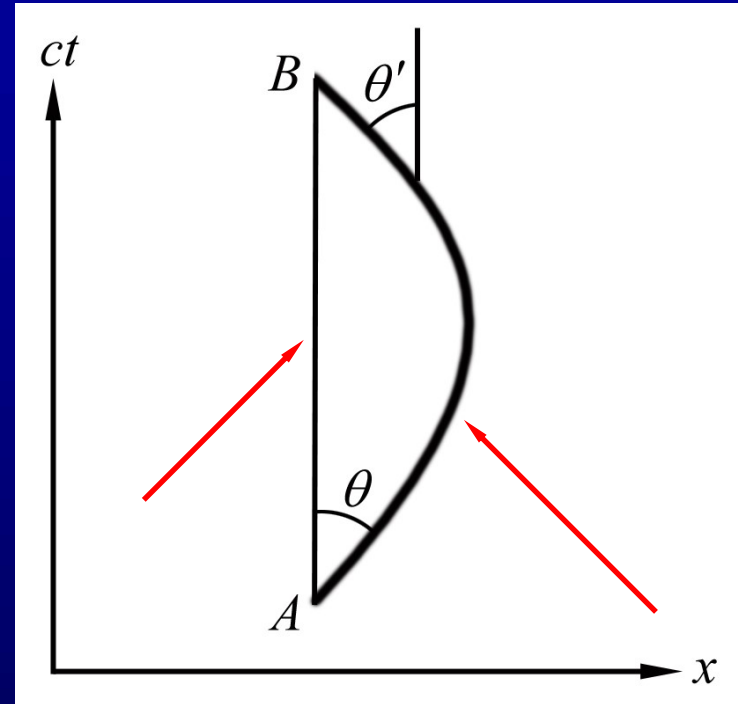
OQ ? Conic surface?

For timelike pair, **proper time** is

$$\Delta \tau = \sqrt{(\Delta t)^2 - \frac{(\Delta x)^2}{c^2}}$$

or

$$\Delta \tau = \int d\tau$$
$$= \int dt \sqrt{1 - \frac{1}{c^2} \left(\frac{dx}{dt} \right)^2}$$

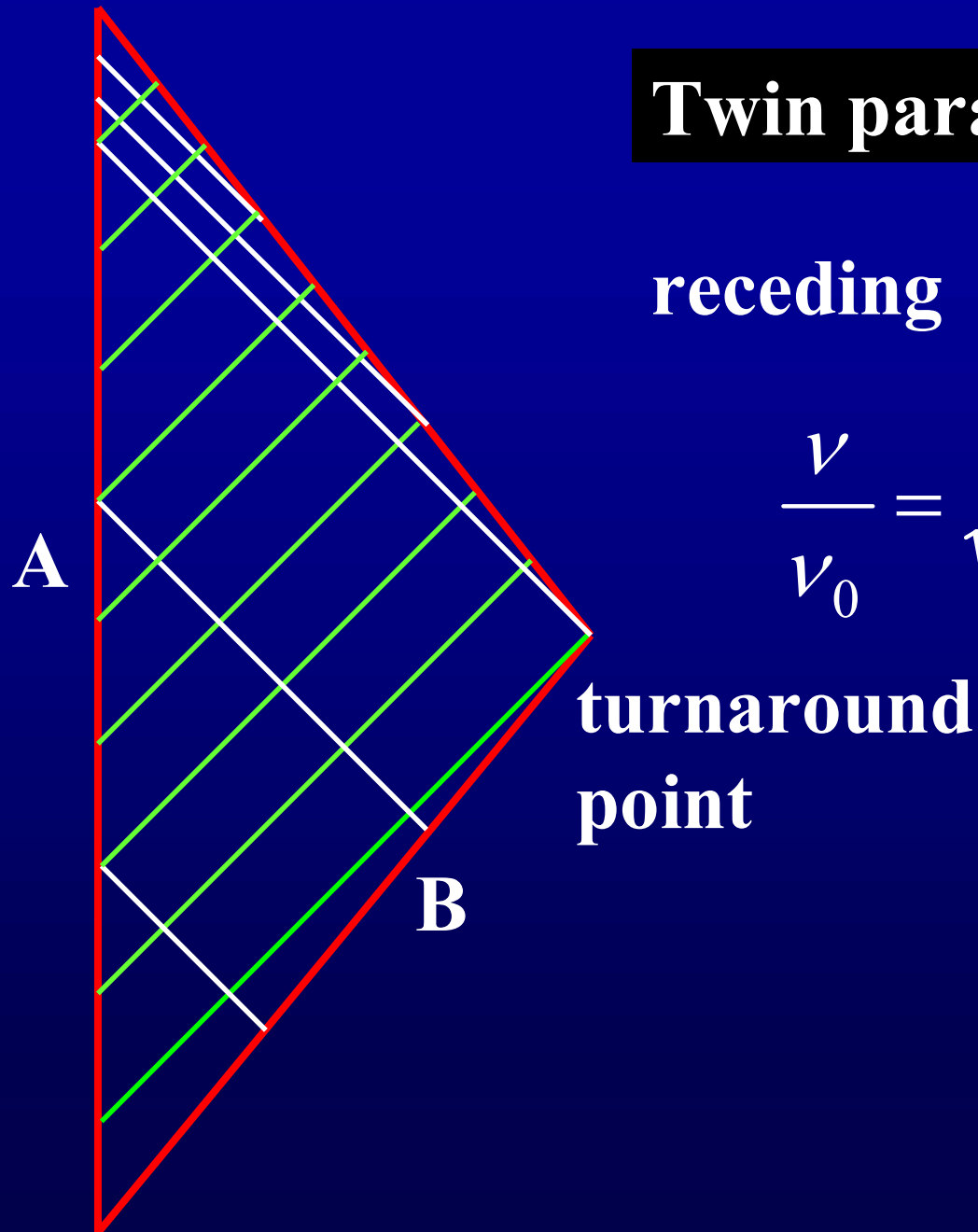


$$\tan \theta = \beta$$

Twin paradox

receding $\beta = 0.8$

$$\frac{v}{v_0} = \sqrt{\frac{1-\beta}{1+\beta}} = \frac{1}{3}$$



A: 10a

B: 6a

The pole-barn paradox :

Suppose a superhero carrying a horizontal pole of 20m running at $0.9c$ towards a barn 10m long with front and rear doors. Can the pole fit in the barn?

$$v = 0.9c$$

$$20 \times \sqrt{1 - \left(\frac{v}{c}\right)^2} = 8.73$$

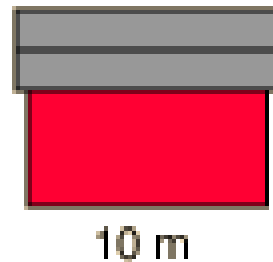
Ground observer:

He can close and then open both doors when the runner and the pole are inside the barn so that the runner and pole are momentarily captured inside the barn and then proceed to exit the barn from the back door.

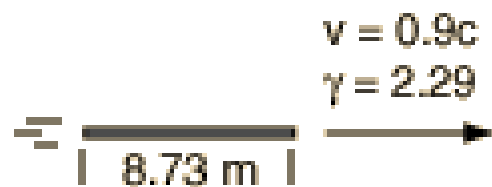
Runner's point of view:

The barn is contracted to 0.44m. The length of the pole is 20m (proper length).

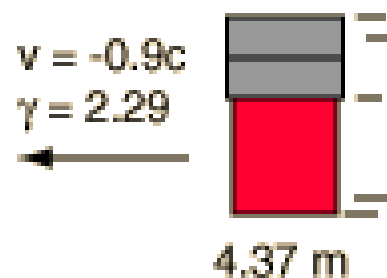
 paradox



A 20 meter pole and a 10 meter barn in their perspective when in the same rest frame.



If the pole has speed $0.9c$, then it is length-contracted by a factor of 2.29 and short enough to fit momentarily within the barn, at least as seen by an observer in the barn.



But from the point of view of the pole, the barn is contracted and the pole will never fit inside it.

The resolution: relativity of simultaneity

The closing of the two doors is measured to be simultaneous by the ground observer. However, since the doors are at different positions, the runner says that they do not close simultaneously. The rear door closes and then opens first, allowing the leading edge of the pole to exit. The front door of the barn does not close until the trailing edge of the pole passes by.

Barn frame of reference

Front of pole enters: $t = 0$

Back of pole enters: $t = 8.73\text{m}/0.9c = 32.29\text{ns}$

Front of pole leaves: $t = 10\text{m}/0.9c = 37.04\text{ns}$

Back of pole leaves: $t = 32.35\text{ns} + 37.04\text{ns}$
 $= 69.38\text{ns}$

The back of the pole enters the barn before the front of the pole leaves, so gates could be closed on both ends, containing the entire pole.

Pole frame of reference

Front of pole enters: $t' = 0$

Front of pole leaves barn: $t' = 4.37\text{m}/0.9c = 16.14\text{ns}$

Back of pole enters: $t' = 20\text{m}/0.9c = 74.07\text{ ns}$

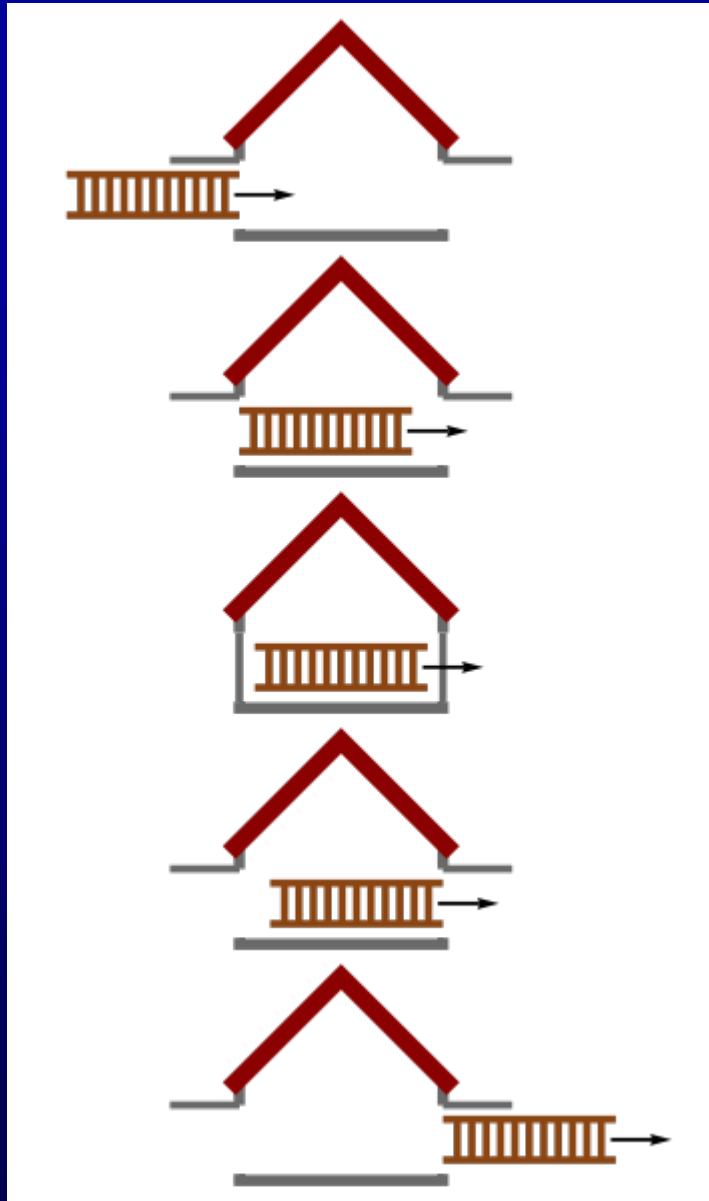
Back of pole leaves: $t' = 16.14\text{ ns} + 74.07\text{ ns}$
 $= 90.21\text{ ns}$

Front gate closes at $t = 32.35\text{ ns}$,

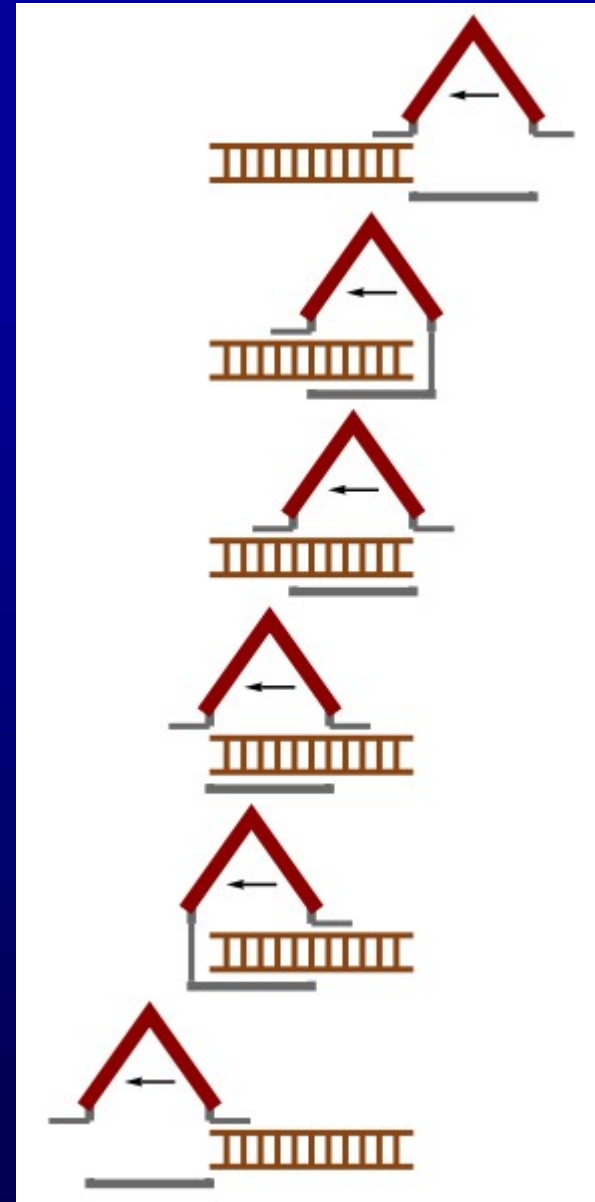
but $t' = \gamma(t - vx/c^2) = 2.29(32.29\text{ ns}) = 74.07\text{ns}$

Back gate closes at $t=32.35\text{ ns}$, but at $x=10\text{m}$. It is simultaneous in the barn frame, but not in the pole frame. The time for back gate closing in the pole frame is

$t' = \gamma(t - vx/c^2) = 2.29(32.35 - (0.9c)(10\text{ m})/c^2) = 5.38\text{ ns}.$



Scenario in the barn frame



Scenario in the pole frame

From the pole point of view, the front gate closes just as the back of the pole enters. The surprising result is that the back gate is seen to close earlier from the pole framework, before the front of the pole reaches it. The gate closings are not simultaneous, and they permit the pole to pass through without hitting either gate.

Pole frame of reference

What happens if the rear door is kept closed and made out of some impenetrable material?

Front of pole touches the back door at :

$$t' = 4.37\text{m}/0.9c = 16.14\text{ns}$$

Assume the shock wave is traveling at speed c .

The information (touching of the door) reaches the back end of the pole at:

$$t' = 16.14\text{ns} + 20\text{m}/c = 82.83$$

It is later than the back of pole enters the barn.

$$(t' = 20\text{m}/0.9c = 74.07 \text{ ns})$$

The Visual Appearance of Rapidly Moving Objects

**W. Greiner, Classical Mechanics Point
Particles and Relativity (Springer, 2004)
p396-399**

9.4 Relativistic kinematics

For close events pair:

$$\left\{ \begin{array}{l} dx' = \gamma (dx - u dt) \\ dy' = dy \\ dz' = dz \\ dt' = \gamma \left(dt - \frac{u}{c^2} dx \right) \\ \quad = \gamma dt \left(1 - v_x \frac{u}{c^2} \right) \end{array} \right.$$

$$\left\{ \begin{array}{l} v'_x = \frac{v_x - u}{1 - v_x \frac{u}{c^2}} \\ v'_y = \frac{v_y}{\gamma \left(1 - v_x \frac{u}{c^2} \right)} \\ v'_z = \frac{v_z}{\gamma \left(1 - v_x \frac{u}{c^2} \right)} \end{array} \right.$$

Example 9.1

Electron-positron collider



Solution:

Lab = S. Take  as S'

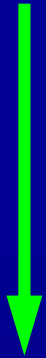
$$u = 0.90c$$

$$v_x = -0.90c$$

$$v'_x = \frac{\overbrace{-0.90c}^{u} - 0.90c}{1 + 0.90^2} = -0.994c$$

Aberration Known:

$$v_z = -c, v_x = v_y = 0, u = v$$

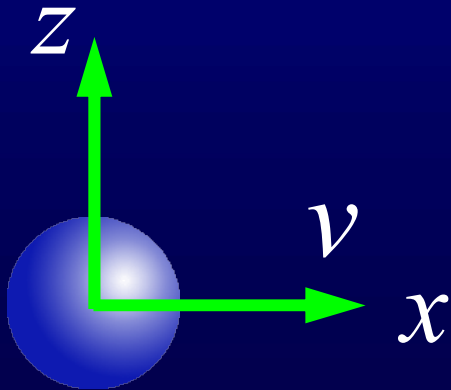


Classically

$$v'_x = v_x - u = -v$$

$$v'_z = v_z = -c$$

$$\tan \alpha' = \frac{v'_x}{v'_z} = \frac{v}{c}$$



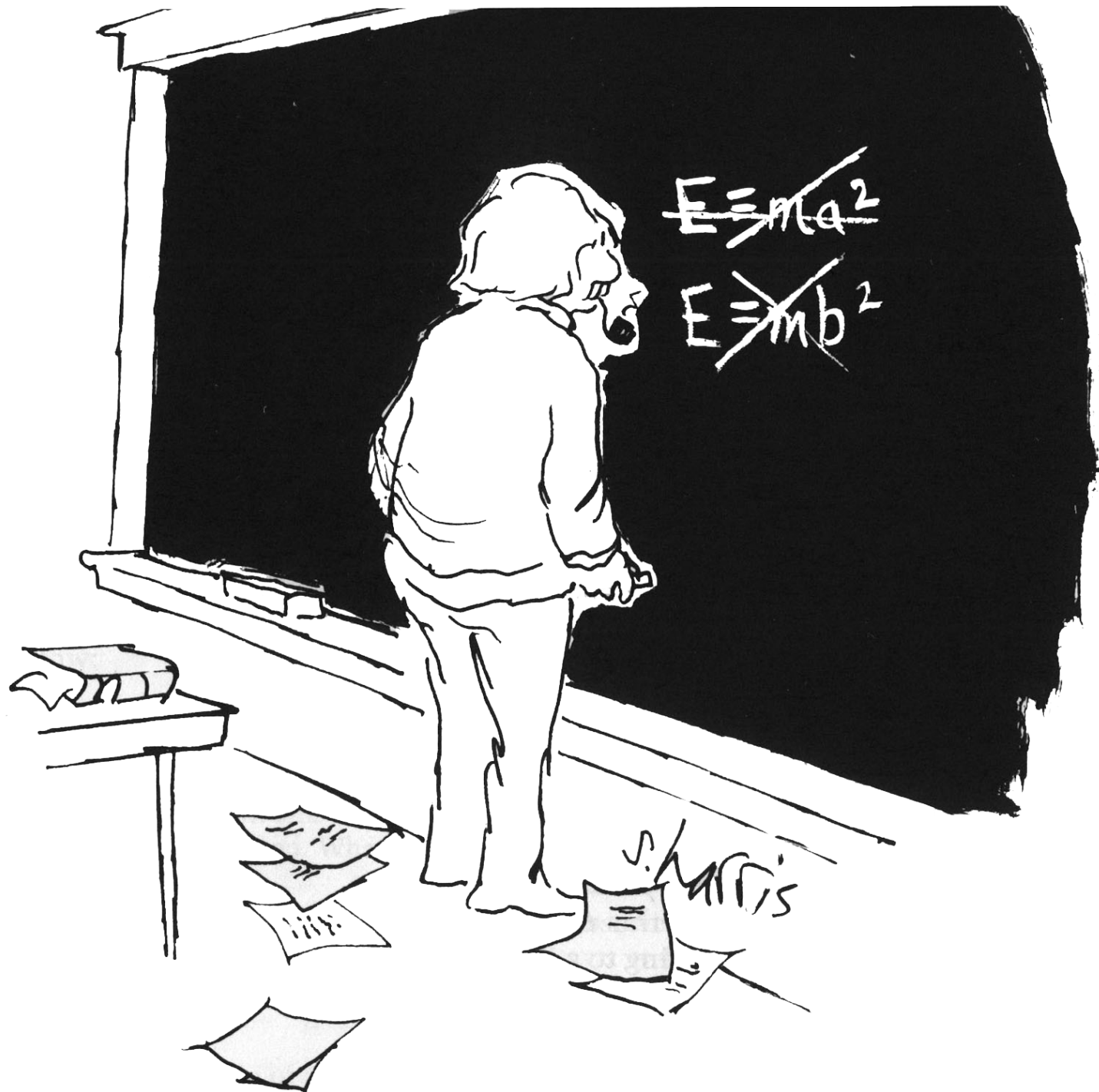
According to special relativity

$$v'_z = \frac{-c}{\gamma \left(1 - \cancel{v}_x \frac{u}{c^2} \right)} \quad v'_x = \frac{-v}{1 - \cancel{\gamma}_x \frac{u}{c^2}}$$

$$\tan \alpha' = \frac{v'_x}{v'_z} = \frac{\gamma v}{c} = \frac{v/c}{\sqrt{1 - (v/c)^2}}$$

$$\sin \alpha' = \frac{v}{c} \quad \text{Stellar aberration} \rightarrow \text{earth's speed}$$

Assignment: 9.13

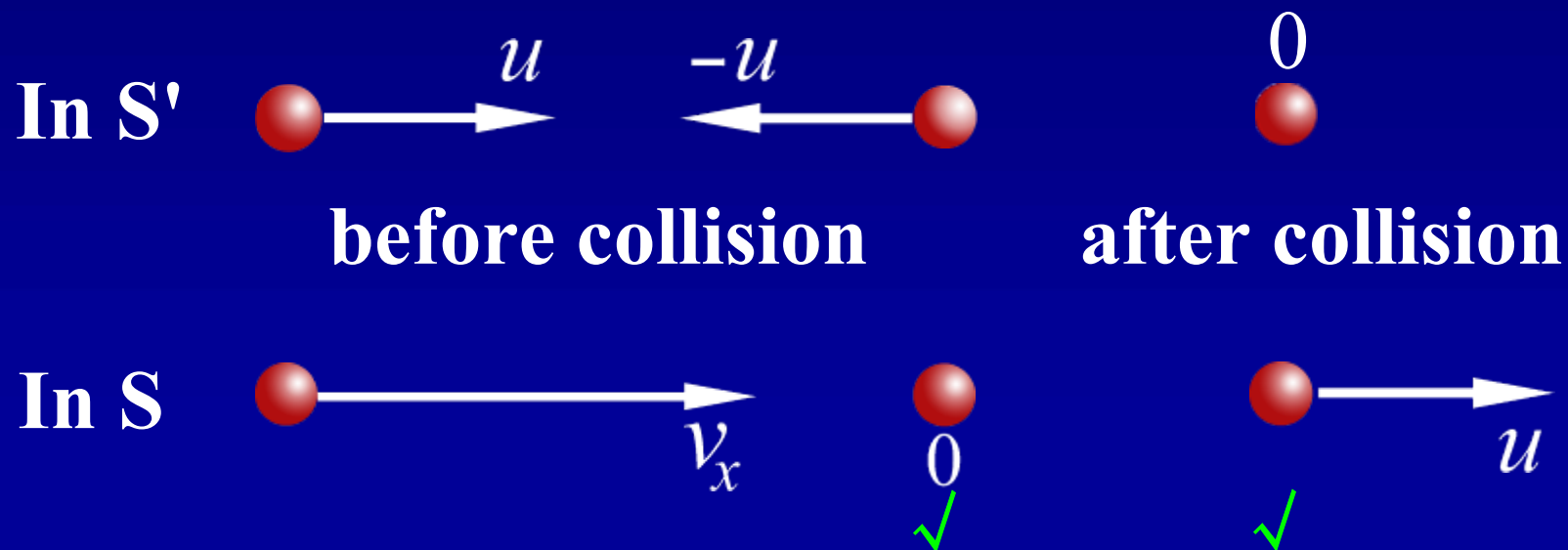


9.5 Relativistic Dynamics

No gravity, no non-inertial frame considered

$$m, p, F, E_k, a$$

Discuss a completely inelastic collision:



In S

$$v_x = \frac{u + u}{1 + u \frac{u}{c^2}}$$

$$v'_x = \frac{v_x - u}{1 - v_x \frac{u}{c^2}}$$

$$m \frac{u + u}{1 + uu/c^2} + m \times 0 \neq 2mu$$

Momentum conservation is “violated”

$$m \equiv m(v),$$

$$m_0 \equiv m(0)$$

rest mass

$$\left\{ \begin{array}{l} m(v_x) + m_0 = m_t(u) \\ \underline{m(v_x) \cdot v_x} + m_0 \cdot 0 = m_t(u)u \\ \phantom{\underline{m(v_x) \cdot v_x} + m_0 \cdot 0 = m_t(u)u} = \underline{[m(v_x) + m_0]}u \end{array} \right.$$

$$m(v_x) = \frac{m_0 u}{v_x - u} = \frac{m_0}{\boxed{\frac{v_x}{u}} - 1}$$

$$v_x = \frac{2u}{1 + \frac{u^2}{c^2}}$$

$$\frac{v_x}{u} = \frac{2}{1 + \left(\frac{v_x}{c}\right)^2 \left(\frac{u}{v_x}\right)^2}$$

i.e.

$$x = \frac{2}{1 + \beta^2/x^2}$$

$$x = \frac{2x^2}{x^2 + \beta^2}$$

$$x - 1 = \sqrt{1 - \beta^2}$$

$$x^2 + \beta^2 = 2x$$

$$\frac{v_x}{u} - 1$$

$$m(v) = \frac{m_0}{\sqrt{1 - \beta^2}} \equiv \gamma m_0 \quad \left(\beta = \frac{v}{c} \right)$$

$$p = mv = \frac{m_0 v}{\sqrt{1 - \frac{v^2}{c^2}}} = \gamma m_0 v$$

$$q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) = \frac{d\mathbf{p}}{dt} \quad \text{If } \mathbf{E} = \mathbf{0} \quad q\mathbf{v} \times \mathbf{B} = \frac{d\mathbf{p}}{dt}$$

$$\begin{aligned} \frac{d(\mathbf{p} \cdot \mathbf{p})}{dt} &= 2\mathbf{p} \cdot \frac{d\mathbf{p}}{dt} = 2\mathbf{p} \cdot (q\mathbf{v} \times \mathbf{B}) = 0 \\ &= 2q\mathbf{B} \cdot (\mathbf{p} \times \mathbf{v}) \end{aligned}$$

p^2 is constant, i.e., thence v , m are constants.

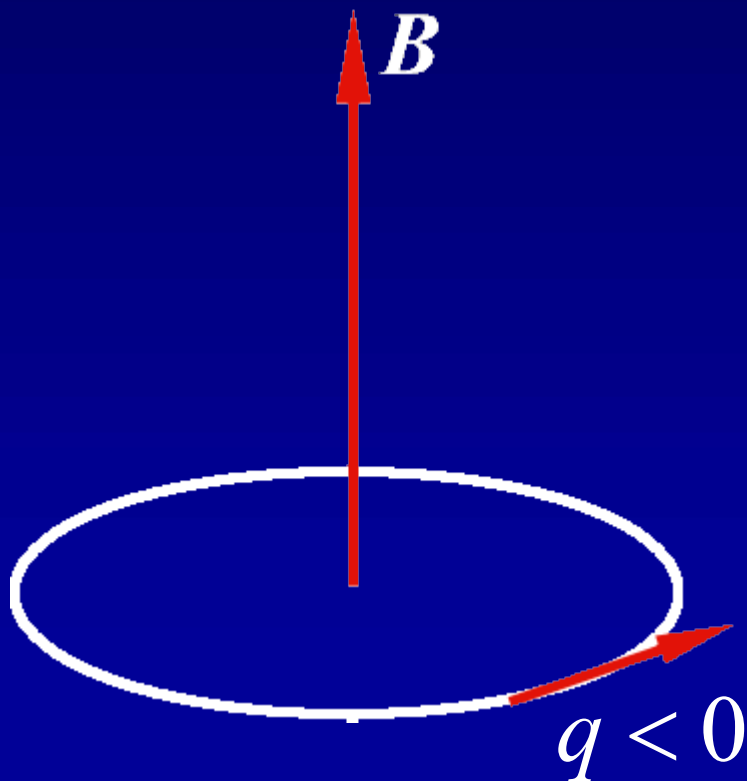
$$q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) = \frac{d\mathbf{p}}{dt} \quad \text{If } \mathbf{E} = \mathbf{0} \quad q\mathbf{v} \times \mathbf{B} = \frac{d\mathbf{p}}{dt}$$

$$m \frac{d\mathbf{v}}{dt} = q\mathbf{v} \times \mathbf{B}$$

$$\mathbf{B} = B\mathbf{k}$$

$$\left\{ \begin{array}{l} \frac{dv_x}{dt} = \frac{qB}{m} v_y \\ \frac{dv_y}{dt} = -\frac{qB}{m} v_x \\ \frac{dv_z}{dt} = 0 \end{array} \right. \rightarrow \left\{ \begin{array}{l} \frac{d^2 v_x}{dt^2} = -\left(\frac{qB}{m}\right)^2 v_x \\ v_y = \frac{m}{qB} \frac{dv_x}{dt} \\ v_z = C \end{array} \right.$$

$$\begin{cases} v_x = v_0 \cos(\omega t + \delta) \\ v_y = -v_0 \operatorname{sgn}(q) \sin(\omega t + \delta) \\ v_z = C \end{cases} \quad \left(\omega = \frac{|q|B}{m} \right)$$



$$m \frac{d\mathbf{v}}{dt} = -q\mathbf{B} \times \mathbf{v}$$

$$\omega = -\frac{qB}{m}$$

For $\mathbf{v} \perp \mathbf{B}$

$$\boxed{\gamma m_0} \frac{v^2}{R} = qvB$$

Kinetic energy

$$E_k = \int \mathbf{F} \cdot d\mathbf{x} = \int \frac{d}{dt}(m\mathbf{v}) \cdot d\mathbf{x} = \int \mathbf{v} \cdot d(m\mathbf{v})$$

In Newtonian mechanics:

$$E_k = \int m\mathbf{v} \cdot d\mathbf{v} = \frac{1}{2}mv^2$$

In special relativity

$$m^2(1 - v^2/c^2) = m_0^2$$

$$E_k = \int (v^2 dm + mv dv) \quad m^2(c^2 - v^2) = m_0^2 c^2$$

$$c^2 2m dm - v^2 2m dm - m^2 2v dv = 0$$

$$c^2 dm - \underbrace{v^2 dm + mv dv}_{m^2 v dv} = 0$$

mass energy-momentum relation

$$\begin{aligned} E^2 &= \cancel{m^2 c^4} - \cancel{m_0^2 c^4} + m_0^2 c^4 = m^2 c^4 \left(1 - \frac{1}{\gamma^2} \right) + m_0^2 c^4 \\ &= m^2 c^4 \beta^2 + m_0^2 c^4 = m^2 c^2 v^2 + m_0^2 c^4 \\ &= c^2 p^2 + m_0^2 c^4 \end{aligned}$$

$$E = \sqrt{c^2 p^2 + m_0^2 c^4}$$

For particle of $m_0 = 0$ $E = cp$, $p = E/c$

$$\text{If } cp \ll m_0 c^2, \quad E = m_0 c^2 \sqrt{1 + \frac{p^2}{m_0^2 c^2}} \approx m_0 c^2 + \frac{p^2}{\underline{2m_0}}$$

$$E_k = \int c^2 dm = mc^2 - m_0 c^2$$

$$= m_0 c^2 \left[\frac{1}{\sqrt{1 - v^2/c^2}} - 1 \right]$$

$$= (\gamma - 1)m_0 c^2$$

$$mc^2 = E_k + m_0 c^2$$

Mass-energy

$$\gamma = \frac{mc^2}{m_0 c^2} = \frac{E_k}{m_0 c^2} + 1$$

$$\beta = \sqrt{1 - \frac{1}{\gamma^2}}$$

m_0c^2 **rest energy (rest mass-energy)**

electron 0.511 MeV

proton 938.3 MeV

neutron 939.6 MeV

$$E = mc^2 = E_k + m_0c^2$$

Example 9.4 A 2.50MeV electron in a magnetic field. The radius of the trajectory is $3.0 \times 10^{-2}\text{m}$. Find its speed and magnetic field.

Solution:

$$\begin{aligned}\gamma &= \frac{mc^2}{m_0c^2} = \frac{E_k}{m_0c^2} + 1 \\ &= \frac{2.50}{0.511} + 1 = 5.89\end{aligned}$$

$$\beta = 0.985$$

$$B = \frac{\gamma m_0 v}{qR}$$

$$= \gamma \frac{m_0 c^2}{q} \frac{\beta}{cR}$$

$$= 5.89 \times 0.511 \times 10^6 \text{ V} \frac{0.985}{3 \times 10^8 \text{ m/s} \times 3.0 \times 10^{-2} \text{ m}}$$

$$= 0.33 \text{ T}$$

$$\Delta E = \Delta mc^2$$

In electron-positron pair annihilation processes

$$e + e^+ \rightarrow 2\gamma \quad E_\gamma \geq 0.511 \text{ MeV}$$

Missing kinetic energy in beta decay process

$$n \rightarrow p + e + (\bar{\nu}_e)$$

$$\boxed{939.6} \boxed{938.3} \boxed{0.511}$$

Ionization of hydrogen

$$\text{H} \rightarrow \text{p} + \text{e}^{-} - Q$$

$$E_{\text{i}} = 13.58 \text{ eV}$$

$$\Delta m = E_{\text{i}} / c^2 = 1.48 \times 10^{-8} \text{ u}$$

$$1 \text{ u} \times c^2 = 931 \text{ MeV}$$

$$1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$$

Example 9.5 Binding energy of deuteron

$$\left(m_n c^2 + \underline{m_p c^2}\right) - m_d c^2 = E_B$$

$$m_n c^2 + \underline{m_{\text{}^1\text{H}} c^2} - m_{\text{}^2\text{H}} c^2$$

$$= (1.00867 + \underline{1.00783} - 2.01410) \text{u} c^2 \times 931 \text{MeV/u}$$

$$= 2.23 \text{MeV}$$

Fission & fusion

$^{235}\text{U} \longrightarrow 2 \text{ product nucleus} + \text{neutron}$



Example 9.6

Sun is losing mass

The solar constant is 1340 W/m^2

$$-\frac{d}{dt}m_{\odot} = -\frac{1}{c^2} \frac{dE}{dt}$$

$$= \frac{1}{c^2} 1340 \text{ W/m}^2 \times 4\pi \times (1.50 \times 10^{11} \text{ m})^2$$

$$= 4.21 \times 10^9 \text{ kg/s}$$

$$\frac{-\frac{d}{dt}m_{\odot}}{m_{\odot}} = \frac{4.21 \times 10^9 \text{ kg/s}}{1.99 \times 10^{30} \text{ kg}} \left(\pi \times 10^7 \text{ s/a} \right)$$

$$= 6.6 \times 10^{-14} / \text{a}$$

$$\frac{-\frac{d}{dt}m_{\odot}}{m_{\odot}} \times 5 \times 10^9 \text{ a} \approx 0.03\%$$

acceleration

$$\mathbf{F} = \frac{d\mathbf{p}}{dt} = \frac{d}{dt}(m\mathbf{v}) = m \frac{d\mathbf{v}}{dt} + \frac{dm}{dt}\mathbf{v}$$

$$= m\mathbf{a} + \frac{1}{c^2} \frac{dE_k}{dt} \mathbf{v}$$

$$= m\mathbf{a} + \frac{1}{c^2} (\mathbf{F} \cdot \mathbf{v}) \mathbf{v}$$

$$\mathbf{a} = \frac{\mathbf{F}}{m} - \frac{\mathbf{F} \cdot \mathbf{v}}{mc^2} \mathbf{v}$$

Example 9.7

The speed of light c as a limiting speed

$$E_k = 30\text{GeV} = 3.0 \times 10^{10} \text{eV}$$

$$\gamma = \frac{E_k}{m_0 c^2} + 1 \approx \frac{E_k}{m_0 c^2} \sim 10^4$$

In Newtonian case $E_k = \frac{1}{2} m c^2 \beta^2$

$$\ln E_k = \ln \left(\frac{1}{2} m c^2 \right) + \ln \beta^2$$

$$\frac{\Delta E_k}{E_k} = 2 \frac{\Delta \beta}{\beta}$$

$$E_k = m_0 c^2 \left(\frac{1}{\sqrt{1 - \beta^2}} - 1 \right)$$

$$\ln E_k = \ln(m_0 c^2) + \ln(\gamma - 1)$$

$$\begin{aligned} \frac{\Delta E_k}{E_k} &= \frac{1}{\gamma - 1} \frac{d\gamma}{d\beta} \frac{\gamma\beta}{\gamma} \frac{\Delta\beta}{\beta} \\ &= \frac{\gamma\beta}{\gamma - 1} \frac{d \ln \gamma}{d\beta} \frac{\Delta\beta}{\beta} \end{aligned}$$

$$E_k = m_0 c^2 \left(\frac{1}{\sqrt{1 - \beta^2}} - 1 \right)$$

$$\ln E_k = \ln(m_0 c^2) + \ln(\gamma - 1)$$

$$\begin{aligned} \frac{\Delta E_k}{E_k} &= \frac{1}{\gamma - 1} \frac{d\gamma}{d\beta} \frac{\gamma\beta}{\gamma} \frac{\Delta\beta}{\beta} \\ &= \frac{\gamma\beta}{\gamma - 1} \left(-\frac{1}{2} \right) \left(\frac{-2\beta}{1 - \beta^2} \right) \frac{\Delta\beta}{\beta} \\ &= \gamma(\gamma + 1) \frac{\Delta\beta}{\beta} \sim \gamma^2 \frac{\Delta\beta}{\beta} \end{aligned}$$

Assignment: 9.16, *9.17, 9.19

Four-vectors and four-momentum

In the Newtonian mechanics

$$\mathbf{r} = (x, y, z) = (x_1, x_2, x_3) = x_1 \mathbf{e}_1 + x_2 \mathbf{e}_2 + x_3 \mathbf{e}_3$$

is a vector in 3 dimensional space. (position vector)

Since time t is a scalar, the velocity which is defined as

$$\mathbf{v} = \frac{d\mathbf{r}}{dt}$$

is also a vector. We then can define another vector called momentum by multiplying $\mathbf{v}$ with another scalar, the mass m

$$\mathbf{p} = m\mathbf{v}$$

Rotational transformation of coordinates

$$\begin{aligned}x'_1 &= u_{11}x_1 + u_{12}x_2 + u_{13}x_3 \\x'_2 &= u_{21}x_1 + u_{22}x_2 + u_{23}x_3 \\x'_3 &= u_{31}x_1 + u_{32}x_2 + u_{33}x_3\end{aligned}$$

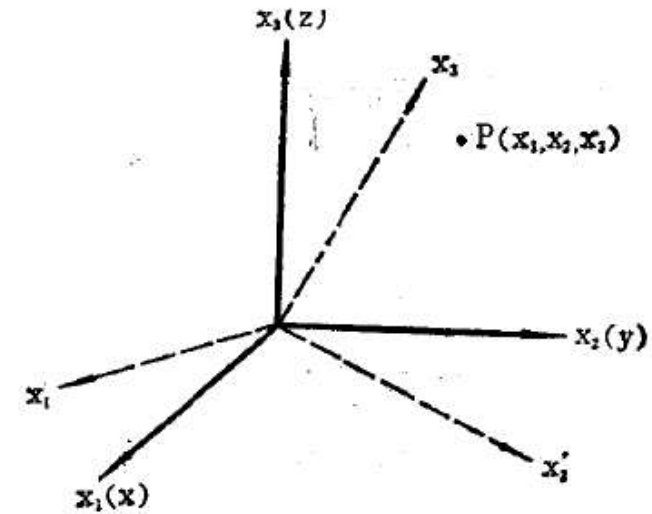
or

$$x'_i = \sum_{j=1}^3 u_{ij} x_j \quad (i = 1, 2, 3).$$

$$x'_i = u_{ij} x_j .$$

summation over repeated indices
Einstein notation

$$\sum_{i=1}^3 u_{ij} u_{ik} = \sum_{i=1}^3 u_{ji} u_{ki} = \delta_{jk}$$



Kronecker delta

$$U = (u_{ij}) \quad \sum_{i=1}^3 u_{ij} u_{ik} = \sum_{i=1}^3 u_{ji} u_{ki} = \delta_{jk}$$

$$UU^T = U^T U = I \quad (U \text{ is an orthogonal matrix})$$

$$\det U = 1 \quad \text{for rotation matrix}$$

$$I \quad \text{identity matrix} \quad \longleftrightarrow \quad E \quad \text{unit matrix (Einheitsmatrix)}$$

If $A = (A_1, A_2, A_3)$ transforms in the same way as position vector r for the rotation of the coordinates, then A is a vector.

$$\begin{array}{lll} r \leftrightarrow -r & A \leftrightarrow -A & \text{polar vector} \\ r \leftrightarrow -r & A \leftrightarrow A & \text{axial (pseudo) vector} \end{array}$$

In the relativistic mechanics, time t is no longer a scalar. We should define a four-vector with time t as one of its components

$$X = (x_0, x_1, x_2, x_3) = (ct, x, y, z) = (ct, \mathbf{r}).$$

The Lorentz transformation of components of X can be expressed as

$$x'_0 = \gamma(x_0 - \beta x_1),$$

$$x'_1 = \gamma(x_1 - \beta x_0),$$

$$x'_2 = x_2,$$

$$x'_3 = x_3.$$

We define the dot product

$$X \cdot X = x_0^2 - x_1^2 - x_2^2 - x_3^2 = (ct)^2 - x^2 - y^2 - z^2.$$

One can see that it is the square of the spacetime interval from the between the origin and point (ct, x, y, z) . And it is unchanged under the Lorentz transformation (Lorentz invariant).

How to define the four momentum?

$$\frac{dr}{dt} \rightarrow \frac{dX}{d\tau} \quad d\tau - \text{proper time interval}$$

$$d\tau = \frac{1}{c} \sqrt{(cdt)^2 - (dx)^2 - (dy)^2 - (dz)^2} = dt \sqrt{1 - \frac{v^2}{c^2}}$$

Therefore, the four-momentum can be defined as

$$\begin{aligned} P &= m_0 \left(\frac{dx_0}{d\tau}, \frac{dx_1}{d\tau}, \frac{dx_2}{d\tau}, \frac{dx_3}{d\tau} \right) \\ &= \left(\frac{m_0 c}{\sqrt{1 - v^2/c^2}}, \frac{m_0}{\sqrt{1 - v^2/c^2}} \frac{d\mathbf{r}}{dt} \right) \\ &= (P_0, \mathbf{p}) = \left(\frac{E}{c}, \mathbf{p} \right), \end{aligned}$$

where $E = mc^2 = \gamma m_0 c^2$, $\mathbf{p} = m\mathbf{v} = \gamma m_0 \mathbf{v}$.

The components of P transform as a four vector as we go from one frame to another:

$$P'_0 = \gamma(P_0 - \beta P_1),$$

$$P'_1 = \gamma(P_1 - \beta P_0),$$

$$P'_2 = P_2,$$

$$P'_3 = P_3.$$

The dot product

$$\begin{aligned} P \cdot P &= P_0^2 - P_1^2 - P_2^2 - P_3^2 \\ &= \gamma^2 m_0^2 c^2 - \gamma^2 m_0^2 v^2 = m_0^2 c^2 \end{aligned}$$

is also Lorentz invariant.

Four-velocity

$$U = \frac{dX}{d\tau} = \left(\frac{dx_0}{d\tau}, \frac{dx_1}{d\tau}, \frac{dx_2}{d\tau}, \frac{dx_3}{d\tau} \right) \\ = (\gamma c, \gamma \mathbf{v}).$$

Four-acceleration

$$A = \frac{dU}{d\tau} = \left(\frac{d\gamma}{d\tau} c, \frac{d(\gamma \mathbf{v})}{d\tau} \right) \\ = \left(\frac{d\gamma}{dt} \frac{dt}{d\tau} c, \frac{d\gamma}{dt} \frac{dt}{d\tau} \mathbf{v} + \gamma \frac{d\mathbf{v}}{dt} \frac{dt}{d\tau} \right) \\ = \gamma \left(\frac{d\gamma}{dt} c, \frac{d\gamma}{dt} \mathbf{v} + \gamma \mathbf{a} \right).$$

Four-force

$$\begin{aligned} F &= \frac{dP}{d\tau} = \frac{d}{d\tau} \left(\frac{E}{c}, \mathbf{p} \right) \\ &= \left(\frac{1}{c} \frac{dE}{dt} \frac{dt}{d\tau}, \frac{d\mathbf{p}}{dt} \frac{dt}{d\tau} \right) \\ &= \gamma \left(\frac{1}{c} \frac{dE}{dt}, \mathbf{f} \right) \end{aligned}$$

Where $\mathbf{f} = \frac{d\mathbf{p}}{dt} = \frac{d(\gamma m_0 \mathbf{v})}{dt}$ is the usual 3-force.

Four-force

$$F = \frac{dP}{d\tau} = \frac{d(m_0 U)}{d\tau} = m_0 \frac{dU}{d\tau} = m_0 A.$$

Using the result

$$A = \gamma \left(\frac{d\gamma}{dt} c, \frac{d\gamma}{dt} \mathbf{v} + \gamma \mathbf{a} \right);$$

$$F = \gamma \left(\frac{1}{c} \frac{dE}{dt}, \mathbf{f} \right),$$

we have

$$\begin{aligned}
 \mathbf{f} &= m_0 \frac{d\gamma}{dt} \mathbf{v} + m_0 \gamma \mathbf{a} = \frac{d(m_0 \gamma)}{dt} \mathbf{v} + m_0 \gamma \mathbf{a} \\
 &= \frac{1}{c^2} \frac{d(mc^2)}{dt} \mathbf{v} + m \mathbf{a} = \frac{1}{c^2} \frac{d(m_0 c^2 + E_k)}{dt} \mathbf{v} + m \mathbf{a} \\
 &= \frac{1}{c^2} (\mathbf{f} \cdot \mathbf{v}) \mathbf{v} + m \mathbf{a}.
 \end{aligned}$$

That is the Eq.(9.5.12) of the textbook.

**David J Morin, Special Relativity For the
Enthusiastic Beginner**